ANew L J Institute of Engineering & Technology

Sem: 1st – CSE/CSE(AIML)/CSE(DS)

Sub: Mathematics-I

Sub Code: BE01R00041

Module 5: Multiple Integral

Weightage: 20%

Topics:

- Double integrals (Cartesian & Polar)
- Evaluation of Double Integral over non rectangular(General) region
- Change of order of integration
- Change of variables $(x, y \to r, \theta)$
- Triple integrals
- Area by double integrals

1. Double Integrals

Integration of function of two variables is called *double integral*. They are useful in evaluating area, volume, surface area, etc.

Propreties of Double Integrals

For $f(x,y)$ and $g(x,y)$ continuous in region R with k as rational number,

(1) $\iint_R [f(x,y) \pm g(x,y)]\,dA = \iint_R f(x,y)\,dA \pm \iint_R g(x,y)\,dA$

(2) Linearity Property: $\iint_R kf(x,y)\,dA = k\iint_R f(x,y)\,dA$, where k is a constant

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(3) Additivity of domain: For $f(x,y)$ continuous in region R, where $R = R_1 \cup R_2$ where R_1 and R_2 are non-overlapping regions whose union is R.

$$\iint_R f(x,y)\,dA = \iint_{R_1} f(x,y)\,dA + \iint_{R_2} f(x,y)\,dA$$

(4) $\iint_R f(x,y)\,dA \geq 0$ if $f(x,y) \geq 0$ on region R

(5) Inequality preserves: $\iint_R f(x,y)\,dA \geq \iint_R g(x,y)\,dA$ if $f(x,y) \geq g(x,y)$ on R

(6) Triangular inequality: $\left|\iint_R f(x,y)\,dA\right| \leq \iint_R |f(x,y)|\,dA$

Double integral in cartesian co-ordinate over rectangle

Double integral of a function $f(x,y)$ over a rectangle region R can be evaluated by two successive single integrations. It has four limits and if all the limits are constant then region of integration R is a rectangle. Here, the function $f(x,y)$ can be integrated w.r.t. any variable first. i.e., $dA = dxdy = dydx$

Therefore,

$$\iint_R f(x,y)\,dA = \int_c^d\int_a^b f(x,y)\,dxdy = \int_a^b\int_c^d f(x,y)\,dydx = \int_c^d\left[\int_a^b f(x,y)\,dx\right]dy = \int_a^b\left[\int_c^d f(x,y)\,dy\right]dx$$

The integral inside the bracket is called "inner integral" which is to be evaluated first and the integral outside the bracket is called "outer integral".

Note: If all the four limits are constant and $f(x,y)$ is explicit then double integral can be written as product of two single integrals.

Example 1: Evaluate $\int_1^2\int_0^1 \frac{1}{(2x+5y)^2}\,dxdy$

Solution:

$$\int_1^2\int_0^1 \frac{1}{(2x+5y)^2}\,dxdy$$

$$= \int_1^2 \left[\frac{(2x+5y)^{-2+1}}{(-2+1)(2)}\right]_0^1 dy$$

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$$=\int_1^2\left[-\frac{1}{2}(2x+5y)^{-1}\right]_0^1 dy$$

$$=-\frac{1}{2}\int_1^2\left[\frac{1}{2+5y}-\frac{1}{5y}\right]dy$$

$$=-\frac{1}{2}\left[\frac{1}{5}\ln(2+5y)-\frac{1}{5}\ln y\right]_1^2$$

$$=-\frac{1}{2}\left[\left(\frac{1}{5}\ln(12)-\frac{1}{5}\ln 2\right)-\left(\frac{1}{5}\ln(7)-\frac{1}{5}\ln 1\right)\right]$$

$$=-\frac{1}{2}\left(\frac{1}{5}\right)[\ln 12-\ln 2-\ln 7]$$

$$=-\frac{1}{10}[\ln 12-\ln 2-\ln 7]$$

Example 2: Evaluate $\int_0^1\int_1^2 xy\,dydx$

Solution:

<u>Method 1</u>

$$\int_0^1\int_1^2 xy\,dydx$$

$$=\int_0^1\left(\int_1^2 xy\,dy\right)dx$$

$$=\int_0^1\left[x\frac{y^2}{2}\right]_1^2 dx$$

$$=\int_0^1\frac{x}{2}[4-1]dx$$

$$=\frac{3}{2}\int_0^1 xdx$$

$$=\frac{3}{2}\left(\frac{x^2}{2}\right)_0^1$$

$$=\frac{3}{4}(1-0)\ =\frac{3}{4}$$

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Method 2

$$\int_0^1\int_1^2 xy\,dydx$$

Since both the limits are constant and integrand (function) is explicit in x and y,

$$\int_0^1 x\,dx\int_1^2 y\,dy$$

$$=\left(\frac{x^2}{2}\right)_0^1\left(\frac{y^2}{2}\right)_1^2$$

$$=\frac{1}{2}\frac{1}{2}(1-0)(4-1)$$

$$=\frac{3}{4}$$

Example 3: Evaluate $\int_{-1}^1\int_0^2\left(1-6x^2y\right)dxdy$

Solution:

$$\int_{-1}^1\int_0^2\left(1-6x^2y\right)dxdy$$

$$=\int_{-1}^1\left[\int_0^2\left(1-6x^2y\right)dx\right]dy$$

$$=\int_{-1}^1\left[x-6y\frac{x^3}{3}\right]_0^2 dy$$

$$=\int_{-1}^1\left[x-2yx^3\right]_0^2 dy$$

$$=\int_{-1}^1\left[2-16y\right]dy$$

$$=\left[2y-16\frac{y^2}{2}\right]_{-1}^1$$

$$=\left[2y-8y^2\right]_{-1}^1$$

$$=2-8+2+8$$

$$=4$$

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Example 4: Calculate $\iint_R f(x,y)dA$ for $f(x,y) = 100 - 6x^2y$
$R: 0 \le x \le 2, -1 \le y \le 1$.

Solution:

$$\int_{-1}^{1}\int_0^2 (100 - 6x^2y)dx\,dy$$

$$= \int_{-1}^{1}\left[\int_0^2 (100 - 6x^2y)dx\right]dy$$

$$= \int_{-1}^{1}\left[100x - 6y\frac{x^3}{3}\right]_0^2 dy$$

$$= \int_{-1}^{1}[100x - 2yx^3]_0^2 dy$$

$$= \int_{-1}^{1}[200 - 16y]dy$$

$$= \left[200y - 16\frac{y^2}{2}\right]_{-1}^{1}$$

$$= [200y - 8y^2]_{-1}^{1}$$

$$= 200 - 8 + 200 + 8$$

$$= 400$$

Example 5: Evaluate $\int_0^1\int_0^1 \frac{1}{\sqrt{(1-x^2)(1-y^2)}}dxdy$

Solution:

$$\int_0^1\int_0^1 \frac{1}{\sqrt{(1-x^2)(1-y^2)}}dxdy$$

$$= \int_0^1\left[\int_0^1\left(\frac{1}{\sqrt{1-x^2}}\right)dx\right]\frac{1}{\sqrt{1-y^2}}dy$$

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$$= \int_0^1 \left[\left(\sin^{-1} \frac{x}{1} \right)_0^1 \right] \frac{1}{\sqrt{1-y^2}} dy$$

$$= \int_0^1 [\sin^{-1} 1 - \sin^{-1} 0] \frac{1}{\sqrt{1-y^2}} dy$$

$$= \frac{\pi}{2} \int_0^1 \frac{1}{\sqrt{1-y^2}} dy$$

$$= \frac{\pi}{2} \left[\left(\sin^{-1} \frac{y}{1} \right)_0^1 \right]$$

$$= \frac{\pi}{2} [\sin^{-1} 1 - \sin^{-1} 0]$$

$$= \frac{\pi^2}{4}$$

Example 6: Evaluate $\int_0^1 \int_0^2 \frac{1}{\sqrt{(4-x^2)(1-y^2)}} dx\, dy$

Solution:

$$= \int_0^1 \left[\int_0^2 \left(\frac{1}{\sqrt{4-x^2}} \right) dx \right] \frac{1}{\sqrt{1-y^2}} dy$$

$$= \int_0^1 \left[\left(\sin^{-1} \frac{x}{2} \right)_0^2 \right] \frac{1}{\sqrt{1-y^2}} dy$$

$$= \int_0^1 [\sin^{-1} 1 - \sin^{-1} 0] \frac{1}{\sqrt{1-y^2}} dy$$

$$= \frac{\pi}{2} \int_0^1 \frac{1}{\sqrt{1-y^2}} dy$$

$$= \frac{\pi}{2} \left[\left(\sin^{-1} \frac{y}{1} \right)_0^1 \right]$$

$$= \frac{\pi}{2} [\sin^{-1} 1 - \sin^{-1} 0]$$

$$= \frac{\pi^2}{4}$$

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Example 7: Evaluate $\int_0^3 \int_0^1 (x^2 + 3y^2)\, dydx$

$$= \int_0^3 \left[\int_0^1 (x^2 + 3y^2) dy \right] dx$$

$$= \int_0^3 \left[\left(x^2 y + 3\frac{y^3}{3} \right)_0^1 \right] dx$$

$$= \int_0^3 [\, x^2 + 1] dx$$

$$= \left[\frac{x^3}{3} + x \right]_0^3$$

$$= 9 + 3$$

$$= 12$$

Example 8: Evaluate $\int_0^2 \int_0^{\frac{\pi}{2}} x\, sinydydx$

$$= \int_0^2 \left[\int_0^{\frac{\pi}{2}} (siny) dy \right] x.dx$$

$$= \int_0^2 \left[(cosy)_0^{\frac{\pi}{2}} \right] x.dx$$

$$= \int_0^2 [\, x] dx = \left[\frac{x^2}{2} \right]_0^2 = 2$$

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Example 9: Evaluate $\int_0^4 \int_0^3 \left(x^2 + y^2 \right) dxdy$

Ans: 100

Example 10: Evaluate $\int_0^3 \int_0^2 (4 - y^2) dydx$

Ans: 16

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❖ Double integral in cartesian co-ordinate over general region :

Double integral of a function $f(x,y)$ over a region R can be evaluated by two successive integrations. There are two different methods to evaluate a double integral.

Method I: Inner limit as a function of *x* (limit of *y*)

Suppose R given by $x=a, x=b, y=f_1(x), y=f_2(x)$ then drawing first lines parallel to y-axis so that $y=f_1(x)$ and $y=f_2(x)$ represent boundary of R. Then

$$\iint_R f(x,y)\,dxdy = \int_{x=a}^{b}\left[\int_{y=f_1(x)}^{f_2(x)} f(x,y)\,dy\right]dx$$

Here integrate first over y (treating x as constant) and then integrating w.r.t x resulting the function of x from a to b.

Method II: Inner limit as a function of *y* (limit of *x*)

Suppose R given by $y=c, y=d, x=f_1(y), x=f_2(y)$ then drawing first lines parallel to x-axis so that $x=f_1(y)$ and $x=f_2(y)$ represent boundary of R. Then

$$\iint_R f(x,y)\,dxdy = \int_{y=c}^{d}\left[\int_{x=f_1(y)}^{f_2(y)} f(x,y)\,dx\right]dy$$

Here integrate first over x (treating y as constant) and then integrating w.r.t y resulting the function of y from c to d.

Example 1: Evaluate $\int_0^1\int_{x^2}^{2-x} dydx$

Solution:

$$\int_0^1\int_{x^2}^{2-x} dydx$$

$$=\int_0^1\left(\int_{x^2}^{2-x} 1\,dy\right)dx$$

$$=\int_0^1 (y)_{x^2}^{2-x}\,dx$$

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$$=\int_0^1\left(2-x-x^2\right)dx$$

$$=\left(2x-\frac{x^2}{2}-\frac{x^3}{3}\right)_0^1$$

$$=2-\frac{1}{2}-\frac{1}{3}\ =\frac{7}{6}$$

Example 2: Evaluate $\int_0^1\int_x^{\sqrt{x}}\left(x^2+y^2\right)dxdy$

Solution:

$$\int_0^1\int_x^{\sqrt{x}}\left(x^2+y^2\right)dxdy$$

$$=\int_0^1\left(\int_x^{\sqrt{x}}\left(x^2+y^2\right)dy\right)dx$$

$$=\int_0^1\left[x^2y+\frac{y^3}{3}\right]_x^{\sqrt{x}}dx$$

$$=\int_0^1\left[x^2\left(\sqrt{x}\right)-x^2\left(x\right)+\frac{1}{3}x^{\frac{3}{2}}-\frac{1}{3}x^3\right]dx$$

$$=\int_0^1\left[x^{\frac{5}{2}}-x^3+\frac{1}{3}x^{\frac{3}{2}}-\frac{1}{3}x^3\right]dx$$

$$=\int_0^1\left[x^{\frac{5}{2}}+\frac{1}{3}x^{\frac{3}{2}}-\frac{4}{3}x^3\right]dx$$

$$=\left[\frac{2}{7}x^{\frac{7}{2}}+\frac{1}{3}\frac{2}{5}x^{\frac{5}{2}}-\frac{4}{3}\frac{1}{4}x^4\right]_0^1$$

$$=\left[\frac{2}{7}(1-0)+\frac{2}{15}(1-0)-\frac{1}{3}(1-0)\right]$$

$$=\frac{2}{7}+\frac{2}{15}-\frac{1}{3}$$

$$=\frac{30+14-35}{105}=\frac{9}{105}$$

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Example 3: Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2}\,dxdy$

Solution:

$$\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2}\,dxdy$$

$$=\int_0^1 \left[\int_0^{\sqrt{1+x^2}} \frac{1}{\left(\sqrt{1+x^2}\right)^2+y^2}\,dy\right]dx$$

$$=\int_0^1 \left(\frac{1}{\sqrt{1+x^2}}\tan^{-1}\frac{y}{\sqrt{1+x^2}}\right)_0^{\sqrt{1+x^2}} dx$$

$$=\int_0^1 \frac{1}{\sqrt{1+x^2}}\left(\tan^{-1}\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}}-\tan^{-1}\frac{0}{\sqrt{1+x^2}}\right)dx$$

$$=\int_0^1 \frac{1}{\sqrt{1+x^2}}\left(\tan^{-1}1-\tan^{-1}0\right)dx$$

$$=\int_0^1 \frac{1}{\sqrt{1+x^2}}\left(\frac{\pi}{4}-0\right)dx$$

$$=\frac{\pi}{4}\left[\log\left(x+\sqrt{1+x^2}\right)\right]_0^1$$

$$=\frac{\pi}{4}\left[\log\left(1+\sqrt{2}\right)-\log 1\right]$$

$$=\frac{\pi}{4}\left[\log\left(1+\sqrt{2}\right)\right]$$

Example 4: Evaluate $\int_0^4 \int_0^{\frac{y^2}{4}} x\,dxdy$

Solution:

$$\int_0^4 \int_0^{\frac{y^2}{4}} x\,dxdy$$

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$$= \int_0^4 \left[\int_0^{\frac{y^2}{4}} x\,dx \right] dy$$

$$= \int_0^4 \left[\frac{x^2}{2} \right]_0^{\frac{y^2}{4}} dy$$

$$= \frac{1}{2} \int_0^4 \left[\frac{y^4}{16} - 0 \right] dy$$

$$= \frac{1}{2} \int_0^4 \frac{y^4}{16} dy$$

$$= \frac{1}{32} \left(\frac{y^5}{5} \right)_0^4$$

$$= \frac{1}{32} \left(\frac{4^5}{5} - 0 \right)$$

$$= \frac{32}{5}$$

Example 5: Evaluate $\int_1^{\log 8} \int_0^{\log y} e^{x+y} dxdy$

Solution:

$$\int_1^{\log 8} \int_0^{\log y} e^{x+y} dxdy$$

$$= \int_1^{\log 8} \left[\int_0^{\log y} e^x e^y dx \right] dy$$

$$= \int_1^{\log 8} e^y \left[e^x \right]_0^{\log y} dy$$

$$= \int_1^{\log 8} e^y \left[e^{\log y} - e^0 \right] dy$$

$$= \int_1^{\log 8} e^y \left[y - 1 \right] dy$$

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$$=\left[(y-1)e^{y}-(1)e^{y}\right]_{1}^{\log 8}$$

$$=\left[(\log 8-1)e^{\log 8}-e^{\log 8}\right]-\left[0\cdot e^{1}-e^{1}\right]$$

$$=\left[(\log 8-1)8-8\right]+e$$

$$=\left[8\log 8-8-8\right]+e$$

$$=8\log 8-16+e$$

Example 6: Evaluation $\int_{0}^{1}\int_{0}^{\sqrt{\frac{1-y^2}{2}}}\frac{1}{\sqrt{1-x^2-y^2}}\,dxdy$

Solution:

$$\int_{0}^{1}\int_{0}^{\sqrt{\frac{1-y^2}{2}}}\frac{1}{\sqrt{1-x^2-y^2}}\,dxdy$$

$$=\int_{0}^{1}\left[\int_{0}^{\sqrt{\frac{1-y^2}{2}}}\frac{1}{\sqrt{\left(1-y^2\right)-x^2}}\,dx\right]dy$$

$$=\int_{0}^{1}\left[\sin^{-1}\frac{x}{\sqrt{1-y^2}}\right]_{0}^{\sqrt{\frac{1-y^2}{2}}}dy$$

$$=\int_{0}^{1}\left[\sin^{-1}\frac{\sqrt{\frac{1-y^2}{2}}}{\sqrt{1-y^2}}-\sin^{-1}0\right]dy$$

$$=\int_{0}^{1}\left[\sin^{-1}\frac{1}{\sqrt{2}}-0\right]dy$$

$$=\int_{0}^{1}\left[\frac{\pi}{4}\right]dy$$

$$=\frac{\pi}{4}\left[y\right]_{0}^{1}$$

$$=\frac{\pi}{4}[1-0]=\frac{\pi}{4}$$

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H.W.

Example 7: Evaluate $\int_0^1 \int_0^x e^{\frac{y}{x}} dydx$

Ans: $\frac{1}{2}(e-1)$

Example 8: Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} \sqrt{a^2-x^2-y^2}\, dxdy$

Ans: $\frac{\pi a^3}{6}$

Example 9: Evaluate $\int_0^1 \int_0^y \sin y^2 dydx$

Ans: $\frac{1}{2}(1-\cos 1)$

2. Double integrals in polar coordinates

The integral $\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r,\theta)\, drd\theta$ represents the polar form of the double integration. This integral is first integrated w.r.t. r keeping θ constant and then the resulting expression is integrated w.r.t. θ.

Example 1: Evaluation $\int_0^{\frac{\pi}{2}} \int_0^{1-\sin\theta} r^2 \cos\theta drd\theta$

Solution:

$$\int_0^{\frac{\pi}{2}} \int_0^{1-\sin\theta} r^2 \cos\theta drd\theta$$

$$= \int_0^{\frac{\pi}{2}} \left[\int_0^{1-\sin\theta} r^2 \cos\theta dr \right] d\theta$$

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$$=\int_0^{\frac{\pi}{2}} \cos\theta \left[\frac{r^3}{3}\right]_0^{1-\sin\theta} d\theta$$

$$=\frac{1}{3}\int_0^{\frac{\pi}{2}} \cos\theta (1-\sin\theta)^3 d\theta$$

$$=-\frac{1}{3}\int_0^{\frac{\pi}{2}} (1-\sin\theta)^3 (-\cos\theta) d\theta$$

$$=-\frac{1}{3}\left[\frac{(1-\sin\theta)^4}{4}\right]_0^{\frac{\pi}{2}} \quad \left(\because \int f^n(x) f'(x) = \frac{f^{n+1}(x)}{n+1}\right)$$

$$=-\frac{1}{3}\left(0-\frac{1}{4}\right)$$

$$=\frac{1}{12}$$

Example 2: Evaluation $\int_0^{\frac{\pi}{4}} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr d\theta$

Solution:

$$\int_0^{\frac{\pi}{4}} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} dr d\theta$$

$$=\int_0^{\frac{\pi}{4}} \left[\int_0^{\sqrt{\cos 2\theta}} \frac{1}{2}\frac{2r}{(1+r^2)^2} dr\right] d\theta$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}} \left[\int_0^{\sqrt{\cos 2\theta}} (1+r^2)^{-2} 2r dr\right] d\theta$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{4}} \left[-(1+r^2)^{-1}\right]_0^{\sqrt{\cos 2\theta}} d\theta \quad \left(\because \int [f(x)]^n f'(x) dx = \frac{f^{n+1}(x)}{n+1}, n \neq -1\right)$$

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$$=-\frac{1}{2}\int_0^{\frac{\pi}{4}}\left[\frac{1}{1+\cos 2\theta}-1\right]d\theta$$

$$=-\frac{1}{2}\int_0^{\frac{\pi}{4}}\left[\frac{1}{2\cos^2\theta}-1\right]d\theta$$

$$=-\frac{1}{2}\int_0^{\frac{\pi}{4}}\left[\frac{1}{2}\sec^2\theta-1\right]d\theta$$

$$=-\frac{1}{2}\left[\frac{1}{2}\tan\theta-\theta\right]_0^{\frac{\pi}{4}}$$

$$=-\frac{1}{2}\left[\frac{1}{2}\tan\frac{\pi}{4}-\frac{\pi}{4}-\frac{1}{2}\tan 0+0\right]$$

$$=-\frac{1}{2}\left[\frac{1}{2}(1)-\frac{\pi}{4}\right]$$

$$=-\frac{1}{2}\left[\frac{2-\pi}{4}\right]$$

$$=\frac{\pi-2}{8}$$

Example 3: Evaluate $\int_0^{\pi}\int_0^{a(1-\cos\theta)} 2\pi r^2\sin\theta\, dr\, d\theta$

Solution:

$$\int_0^{\pi}\int_0^{a(1-\cos\theta)} 2\pi r^2\sin\theta\, dr\, d\theta$$

$$=\int_0^{\pi}\left[\int_0^{a(1-\cos\theta)} 2\pi r^2\sin\theta\, dr\right]d\theta$$

$$=\int_0^{\pi} 2\pi\sin\theta\left[\frac{r^3}{3}\right]_0^{a(1-\cos\theta)} d\theta$$

$$=\frac{2\pi}{3}\int_0^{\pi}\sin\theta\left[a^3(1-\cos\theta)^3-0\right]d\theta$$

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$$= \frac{2\pi a^3}{3}\int_0^{\pi}(1-\cos\theta)^3 \sin\theta d\theta$$

$$= \frac{2\pi a^3}{3}\left[\frac{(1-\cos\theta)^4}{4}\right]_0^{\pi}$$

$$= \frac{2\pi a^3}{12}\left[(1-\cos\pi)^4-(1-\cos 0)^4\right]$$

$$= \frac{2\pi a^3}{12}\left[(2)^4-(0)^4\right]$$

$$= \frac{8\pi a^3}{3}$$

H.W.

Example 4: Evaluate $\int_0^{\frac{\pi}{2}}\int_0^{a(1+\sin\theta)} r^2\cos\theta\, dr d\theta$

Ans: $\frac{5}{4}a^3$

Example 5: Evaluate $\int_0^{\frac{\pi}{2}}\int_0^{2a\cos\theta} r^2\sin\theta\, d\theta dr$

Ans: $\frac{2a^3}{3}$

3. Triple integral

Integration of function of three variables is called *triple integral*.

Triple integral in cartesian co-ordinate

Triple integral of a continuous function $f(x,y)$ over a region V can be evaluated by three successive single integrations.

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Let the region V be bounded below by a surface $z = z_1(x, y)$ and above by a surface $z = z_2(x, y)$. Let the projection of region V in $xy-$plane be R which be bounded by the curves $y = y_1(x), y = y_2(x)$ and $x = a, x = b$. Then the triple integral is defined as

$$I = \int_a^b \left[\int_{y_1(x)}^{y_2(x)} \left\{ \int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z)\, dz \right\} dy \right] dx$$

Note:

If all the limits are constant and $f(x, y, z)$ is explicit then triple integral can be written as product of three single integrals.

Example 1: Evaluate $\int_1^2 \int_2^3 \int_0^1 xyz\, dzdydx$

Solution:

Method-I

$$\int_1^2 \int_2^3 \int_0^1 xyz\, dzdydx$$

$$= \int_1^2 \int_2^3 \left[\int_0^1 xyz\, dz \right] dydx$$

$$= \int_1^2 \int_2^3 xy \left[\frac{z^2}{2} \right]_0^1 dydx$$

$$= \frac{1}{2} \int_1^2 \int_2^3 xy[1-0]\, dydx$$

$$= \frac{1}{2} \int_1^2 \left[\int_2^3 xydy \right] dx$$

$$= \frac{1}{2} \int_1^2 x \left[\frac{y^2}{2} \right]_2^3 dx$$

$$= \frac{1}{4} \int_1^2 x[9-4]\, dx$$

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$$= \frac{5}{4}\int_1^2 x dx$$

$$= \frac{5}{4}\left[\frac{x^2}{2}\right]_1^2$$

$$= \frac{5}{8}[4-1]$$

$$= \frac{15}{8}$$

Method-II

Since all the limits are constant and integrand (function) is explicit in x, y and z, the integral can be written as

$$\int_1^2 x\,dx \int_2^3 y\,dy \int_0^1 z\,dz$$

$$= \left(\frac{x^2}{2}\right)_1^2 \left(\frac{y^2}{2}\right)_2^3 \left(\frac{z^2}{2}\right)_0^1$$

$$= \frac{1}{8}(4-1)(9-4)(1-0)$$

$$= \frac{1}{8}(3)(5)(1)$$

$$= \frac{15}{8}$$

Example 2: Evaluate $\int_0^2\int_1^3\int_1^2 xy^2 z\,dzdxdy$

Solution:

$$\int_0^2\int_1^3\int_1^2 xy^2 z\,dzdxdy$$

$$= \int_0^2\int_1^3 \left[\int_1^2 xy^2 z\,dz\right] dxdy$$

$$= \int_0^2\int_1^3 xy^2 \left[\frac{z^2}{2}\right]_1^2 dxdy$$

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$$=\frac{1}{2}\int_0^2\int_1^3 xy^2\left[4-1\right]dxdy$$

$$=\frac{3}{2}\int_0^2\left[\int_1^3 xy^2dx\right]dy$$

$$=\frac{3}{2}\int_0^2 y^2\left[\frac{x^2}{2}\right]_1^3 dy$$

$$=\frac{3}{4}\int_0^2 y^2\left[9-1\right]dy$$

$$=\frac{3\times 8}{4}\int_0^2 y^2dy$$

$$=6\left[\frac{y^3}{3}\right]_0^2$$

$$=2\left[8-0\right]$$

$$=16$$

Example 3: Evaluate $\int_0^1\int_0^\pi\int_0^\pi y\sin z\,dxdydz$

Solution:

$$\int_0^1\int_0^\pi\int_0^\pi y\sin z\,dxdydz$$

$$=\int_0^1\int_0^\pi\left[\int_0^\pi y\sin z\,dx\right]dydz$$

$$=\int_0^1\int_0^\pi y\sin z\left[x\right]_0^\pi dydz$$

$$=\int_0^1\int_0^\pi y\sin z\left[\pi-0\right]dydz$$

$$=\pi\int_0^1\left[\int_0^\pi y\sin zdy\right]dz$$

$$=\pi\int_0^1\sin z\left[\frac{y^2}{2}\right]_0^\pi dz$$

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$$= \frac{\pi}{2}\int_0^1 \sin z\left[\pi^2 - 0\right]dz$$

$$= \frac{\pi^3}{2}\int_0^1 \sin z dz$$

$$= \frac{\pi^3}{2}\left[-\cos z\right]_0^1$$

$$= \frac{\pi^3}{2}\left[-\left(\cos 1 - \cos 0\right)\right]$$

$$= \frac{\pi^3}{2}\left(1 - \cos 1\right)$$

Example 4: Evaluate $\int_0^1\int_0^x\int_0^{\sqrt{x+y}} z\, dxdydz$

Solution:

$$\int_0^1\int_0^x\int_0^{\sqrt{x+y}} z\, dxdydz$$

$$= \int_0^1\int_0^x\left[\int_0^{\sqrt{x+y}} z\, dz\right]dydx$$

$$= \int_0^1\int_0^x\left[\frac{z^2}{2}\right]_0^{\sqrt{x+y}} dydx$$

$$= \int_0^1\int_0^x\left[\frac{x+y}{2} - 0\right]dydx$$

$$= \frac{1}{2}\int_0^1\left[\int_0^x (x+y)\, dy\right]dx$$

$$= \frac{1}{2}\int_0^1\left[xy + \frac{y^2}{2}\right]_0^x dx$$

$$= \frac{1}{2}\int_0^1\left[x^2 + \frac{x^2}{2}\right]dx$$

$$= \frac{1}{2}\int_0^1\left[\frac{3}{2}x^2\right]dx$$

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$$=\frac{3}{4}\int_0^1 x^2 dx$$

$$=\frac{3}{4}\left[\frac{x^3}{3}\right]_0^1$$

$$=\frac{3}{4}\left[\frac{1}{3}-0\right]$$

$$=\frac{1}{4}$$

Example 5: Evaluate $\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} xyz\, dxdydz$

Solution:

$$\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} xyz\, dxdydz$$

$$=\int_0^1\int_0^{\sqrt{1-x^2}}\left[\int_0^{\sqrt{1-x^2-y^2}} xyz\, dz\right]dydx$$

$$=\int_0^1\int_0^{\sqrt{1-x^2}} xy\left[\frac{z^2}{2}\right]_0^{\sqrt{1-x^2-y^2}} dydx$$

$$=\frac{1}{2}\int_0^1\int_0^{\sqrt{1-x^2}} xy\left[1-x^2-y^2-0\right]dydx$$

$$=\frac{1}{2}\int_0^1\left[\int_0^{\sqrt{1-x^2}}\left(xy-x^3y-xy^3\right)dy\right]dx$$

$$=\frac{1}{2}\int_0^1\left[x\frac{y^2}{2}-x^3\frac{y^2}{2}-x\frac{y^4}{4}\right]_0^{\sqrt{1-x^2}} dx$$

$$=\frac{1}{8}\int_0^1\left[2x\left(1-x^2\right)-2x^3\left(1-x^2\right)-x\left(1-x^2\right)^2-0\right]dx$$

$$=\frac{1}{8}\int_0^1\left[2x-2x^3-2x^3+2x^5-x\left(1-2x^2+x^4\right)\right]dx$$

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$$=\frac{1}{8}\int_0^1\left[2x-2x^3-2x^3+2x^5-x+2x^3-x^5\right]dx$$

$$=\frac{1}{8}\int_0^1\left[x^5-2x^3+x\right]dx$$

$$=\frac{1}{8}\left[\frac{x^6}{6}-2\frac{x^4}{4}+\frac{x^2}{2}\right]_0^1$$

$$=\frac{1}{8}\left[\frac{1}{6}-\frac{1}{2}+\frac{1}{2}-0\right]$$

$$=\frac{1}{8}\left[\frac{1}{6}\right]$$

$$=\frac{1}{48}$$

Example 6: Evaluate $\int_0^2\int_1^2\int_0^{yz} xyz\,dxdydz$

Solution:

$$\int_0^2\int_1^2\int_0^{yz} xyz\,dxdydz$$

$$=\int_0^2\int_1^2\left[\int_0^{yz} xyz\,dx\right]dydz$$

$$=\int_0^2\int_1^2 yz\left[\frac{x^2}{2}\right]_0^{yz}dydz$$

$$=\int_0^2\int_1^2 yz\left[\frac{y^2z^2}{2}-0\right]dydz$$

$$=\frac{1}{2}\int_0^2\left[\int_1^2 y^3z^3dy\right]dz$$

$$=\frac{1}{2}\int_0^2 z^3\left[\frac{y^4}{4}\right]_1^2 dz$$

$$=\frac{1}{8}\int_0^2 z^3\left[2^4-1^4\right]dz$$

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$$= \frac{15}{8}\int_0^2 z^3 dz$$

$$= \frac{15}{8}\left[\frac{z^4}{4}\right]_0^2$$

$$= \frac{15}{32}\left[2^4 - 0\right]$$

$$= \frac{15}{32}[16]$$

$$= \frac{15}{2}$$

Example 7: Evaluate $\int_0^2 \int_1^z \int_0^{yz} xyzdx\, dy\, dz$

Solution:

$$\int_0^2 \int_1^z \int_0^{yz} xyzdx\, dy\, dz$$

$$= \int_0^2 \int_1^z \left[\int_0^{yz} xyzdx\right] dy\, dz$$

$$= \int_0^2 \int_1^z yz\left[\frac{x^2}{2}\right]_0^{yz} dy\, dz$$

$$= \int_0^2 \int_1^z yz\left[\frac{y^2z^2}{2} - 0\right] dy\, dz$$

$$= \frac{1}{2}\int_0^2 \left[\int_1^z y^3z^3 dy\right] dz$$

$$= \frac{1}{2}\int_0^2 z^3\left[\frac{y^4}{4}\right]_1^z dz$$

$$= \frac{1}{8}\int_0^2 z^3[z^4 - 1^4]dz$$

$$= \frac{1}{8}\int_0^2 [z^7 - 1^3]dz$$

$$= \frac{1}{8}\left[\frac{z^8}{8} - \frac{z^4}{4}\right]_0^2$$

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$$= \frac{1}{8}(32-4)$$

$$= \frac{28}{8}$$

$$= \frac{7}{2}$$

Example 8: Evaluation $\int_0^1 \int_0^{\sqrt{z}} \int_0^{2\pi} \left(r^2 \cos^2\theta + z^2\right) r\, d\theta dr dz$

Solution:

$$\int_0^1 \int_0^{\sqrt{z}} \int_0^{2\pi} \left(r^2 \cos^2\theta + z^2\right) r\, d\theta dr dz$$

$$= \int_0^1 \int_0^{\sqrt{z}} \left[\int_0^{2\pi} \left(r^2 \left\{ \frac{1+\cos 2\theta}{2} \right\} + z^2 \right) d\theta \right] r\, dr dz$$

$$= \int_0^1 \int_0^{\sqrt{z}} \left[\frac{r^2}{2}\left(\theta + \frac{\sin 2\theta}{2} \right) + z^2\theta \right]_0^{2\pi} r\, dr dz$$

$$= \int_0^1 \int_0^{\sqrt{z}} \left[\left\{ \frac{r^2}{2}\left(2\pi + \frac{\sin 4\pi}{2} \right) + 2z^2\pi \right\} - \{0\} \right] r\, dr dz$$

$$= \int_0^1 \int_0^{\sqrt{z}} \left[\left\{ \frac{r^2}{2}(2\pi + 0) + 2\pi z^2 \right\} \right] r\, dr dz$$

$$= \int_0^1 \int_0^{\sqrt{z}} \left[\pi r^3 + 2\pi r z^2 \right] dr dz$$

$$= \int_0^1 \left[\pi \frac{r^4}{4} + 2\pi \frac{r^2}{2} z^2 \right]_0^{\sqrt{z}} dz$$

$$= \int_0^1 \left[\pi \frac{z^2}{4} + \pi z z^2 \right] dz$$

$$= \pi \int_0^1 \left[\frac{z^2}{4} + z^3 \right] dz$$

$$= \pi \left[\frac{z^3}{4\times 3} + \frac{z^4}{4} \right]_0^1$$

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$$= \pi\left[\frac{1}{12}+\frac{1}{4}-0\right]$$

$$= \frac{\pi}{12}[1+3]$$

$$= \frac{\pi}{3}$$

H.W.

Example 9: Evaluate $\int_0^1\int_0^1\int_0^e dydxdz$

Ans: $2e$

Example 10: Evaluate $\int_{-1}^{1}\int_0^2\int_0^1 (xz-y^3)dz\,dy\,dx$

Ans: -8

Example 11: Evaluate $\int_0^1\int_0^{1-y}\int_0^{1-y-z} z\,dxdzdy$

Ans: $\frac{1}{24}$

Example 12: Evaluate $\int_0^1\int_0^{1-x}\int_0^{x+y} e^z dxdydz$

Ans: $\frac{1}{2}$

Example 13: Evaluate $\int_1^e\int_1^{\log y}\int_1^{e^x} \log z\,dxdydz$

Ans: $\frac{e^2}{4}-2e+\frac{13}{4}$

Example 14: Evaluate the triple integral $\iiint_B xyz^2 dV$ where B is the rectangular box given by $B=\{(x,y,z): 0\le x\le 1, -1\le y\le 2, 0\le z\le 3\}$

Ans: $\frac{27}{4}$

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4. Evaluation of double integral

➢ When region is rectangle:

If $f(x,y)$ is continuous function on R; $a \le x \le b, c \le y \le d$ then

$$\iint_R f(x,y)\,dA = \int_c^d \int_a^b f(x,y)\,dxdy = \int_a^b \int_c^d f(x,y)\,dydx$$

Example 1: Evaluate $\iint_R 7xy^2 dA, R$ is region bounded by $1 \le x \le 2, 2 \le y \le 3$.

Solution: Region R is bounded by $1 \le x \le 2, 2 \le y \le 3$

Therefore,

$$\iint_R 7xy^2 dA$$

$$= \int_2^3 \left[\int_1^2 7xy^2 dx \right] dy$$

$$= \int_2^3 7y^2 \left[\frac{x^2}{2} \right]_1^2 dy$$

$$= \int_2^3 \frac{7}{2} y^2 [4-1] dy$$

$$= \frac{21}{2} \int_2^3 y^2 dy$$

$$= \frac{21}{2} \left[\frac{y^3}{3} \right]_2^3$$

$$= \frac{7}{2}[27-8]$$

$$= \frac{133}{2}$$

➢ Working rules for evaluation of double integrals over a general region (Non rectangular region):

(1) If the region is bounded by more than one curve then find the points of intersection of all the curves.

(2) Draw all the curves and mark their point of intersection.

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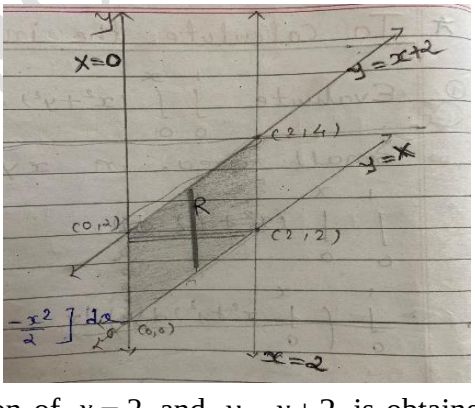
(3) Identify the region of intersection.
(4) Draw a vertical or horizontal strip in the region whichever makes the integration easier.
(5) The vertical strip starts from the lowest part of the region and terminates on the highest part of the region.
(6) For vertical strip:
   i. The lower limit of y is obtained from the curve where the vertical strip starts and the upper limit of y is obtained from the curve where it terminates.
   ii. The lower limit of x is obtained from the leftmost point of the region and the upper limit of x is obtained from the rightmost point of the region.
(7) The horizontal strip starts from the left part of the region and terminates on the right part of the region.
(8) For horizontal strip:
   i. The lower limit of x is obtained from the curve where the horizontal strip starts and the upper limit is obtained from the curve where it terminates.
   ii. The lower limit of y is obtained from the lowest point of the region and the upper limit of y is obtained from the rightmost point of the region.
(9) If variation along the strip changes within the region, then the region is divided into parts.

Example 1: Evaluate $\iint_R (x+y)\,dydx; R$ is the region bounded by

$x=0, x=2, y=x, y=x+2$

Solution:



→ The point of intersection of $x=2$ and $y=x+2$ is obtained as $y=4$. Therefore, the point of intersection is $(2,4)$.

→ The integration can be done w.r.t any variable first. Draw a vertical strip parallel to y-axis which starts from the line $y=x$ and terminates on the line $y=x+2$.

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$\rightarrow$ Limit of $y: y = x$ to $y = x+2$

$\rightarrow$ Limit of $x: x = 0$ to $x = 2$

$$\iint_R (x+y)\,dydx$$

$$= \int_0^2 \int_x^{x+2} (x+y)\,dydx$$

$$= \int_0^2 \left[\int_x^{x+2} (x+y)\,dy \right] dx$$

$$= \int_0^2 \left[xy + \frac{y^2}{2} \right]_x^{x+2} dx$$

$$= \int_0^2 \left[x(x+2) + \frac{(x+2)^2}{2} - x(x) - \frac{x^2}{2} \right] dx$$

$$= \int_0^2 \left[x^2 + 2x + \frac{x^2+4x+4}{2} - x^2 - \frac{x^2}{2} \right] dx$$

$$= \int_0^2 \left[\frac{2x^2+4x+x^2+4x+4-2x^2-x^2}{2} \right] dx$$

$$= \int_0^2 \left[\frac{8x+4}{2} \right] dx$$

$$= \int_0^2 [4x+2]\,dx$$

$$= \left[4\frac{x^2}{2} + 2x \right]_0^2$$

$$= 2(4) + 2(2) - 0$$

$$= 8 + 4$$

$$= 12$$

Example 2: Evaluate $\iint_R e^{2x+3y}\,dA$ where R is the triangle bounded by $x = 0, y = 0$ and $x + y = 1$.

Solution:

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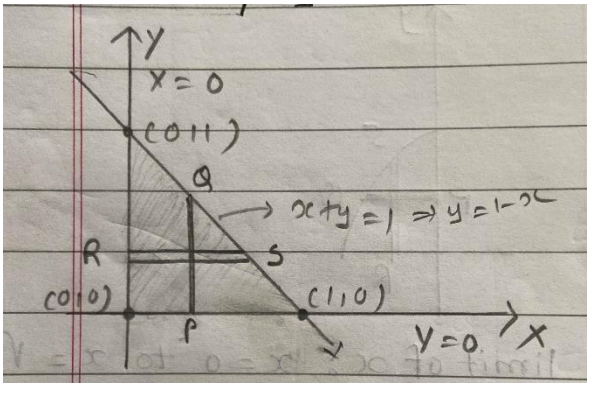

→ The integration can be done w.r.t. any variable first. Draw a vertical strip parallel to y-axis which starts from x-axis i.e. $y=0$ and terminates on the line $x+y=1$.

→ Limit of $y: y=0$ to $y=1-x$

→ Limit of $x: x=0$ to $x=1$

$$I=\int_0^1\int_0^{1-x} e^{2x+3y}\,dy\,dx$$

$$=\int_0^1\left[\int_0^{1-x} e^{2x+3y}\,dy\right]dx$$

$$=\int_0^1\left[\frac{e^{2x+3y}}{3}\right]_0^{1-x}dx$$

$$=\frac{1}{3}\int_0^1\left[e^{2x+3(1-x)}-e^{2x+3(0)}\right]dx$$

$$=\frac{1}{3}\int_0^1\left[e^{3-x}-e^{2x}\right]dx$$

$$=\frac{1}{3}\left[\frac{e^{3-x}}{-1}-\frac{e^{2x}}{2}\right]_0^1$$

$$=\frac{1}{3}\left[\left(\frac{e^2}{-1}-\frac{e^2}{2}\right)-\left(\frac{e^3}{-1}-\frac{e^0}{2}\right)\right]$$

$$=\frac{1}{3}\left[-e^2-\frac{e^2}{2}+e^3+\frac{1}{2}\right]$$

$$=\frac{1}{6}\left[-2e^2-e^2+2e^3+1\right]=\frac{1}{6}\left[-3e^2+2e^3+1\right]$$

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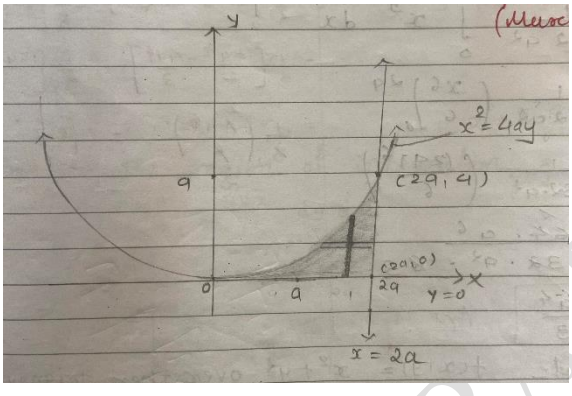
Example 3: Evaluate $\iint_R xy\,dA$, where R is the region bounded by x-axis, the line $x = 2a$ and the parabola $x^2 = 4ay$.

Solution:



→ Point of intersection of the line $x = 2a$ and the parabola $x^2 = 4ay$ is obtained as $4a^2 = 4ay \Rightarrow y = a$. Therefore, the point of intersection is $(2a, a)$.

→ Draw vertical strip parallel to y-axis which starts from x-axis i.e. $y = 0$ and terminate on the parabola $x^2 = 4ay$.

→ Limit of $y : y = 0$ to $y = \dfrac{x^2}{4a}$

→ Limit of $x : x = 0$ to $x = 2a$

$$I = \iint_R xy\,dA$$

$$= \int_0^{2a} \int_0^{\frac{x^2}{4a}} xy\,dy\,dx$$

$$= \int_0^{2a} \left[\int_0^{\frac{x^2}{4a}} xy\,dy \right] dx$$

$$= \int_0^{2a} x \left[\frac{y^2}{2} \right]_0^{\frac{x^2}{4a}} dx$$

$$= \frac{1}{2} \int_0^{2a} x \left[\frac{x^4}{16a^2} - 0 \right] dx$$

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$$= \frac{1}{32a^2}\int_0^{2a} x^5 dx$$

$$= \frac{1}{32a^2}\left[\frac{x^6}{6}\right]_0^{2a}$$

$$= \frac{1}{32\times 6a^2}\left[(2a)^6 - 0\right]$$

$$= \frac{1}{32\times 6a^2}\left(64a^6\right)$$

$$= \frac{a^4}{3}$$

Example 4: Sketch the region of intersection and evaluate $\int_1^4\int_0^{\sqrt{x}} \frac{3}{2}e^{\frac{y}{\sqrt{x}}} dydx$

Solution:

→ Since the inner limits depend on x, the function is integrated first w.r.t. y and then w.r.t. x.

→ Limit of $y: y=0$ to $y=\sqrt{x}$

→ Limit of $x: x=1$ to $x=4$

→ The region is bounded by the line $x=1, y=0, x=4$ and parabola $y^2=x$.

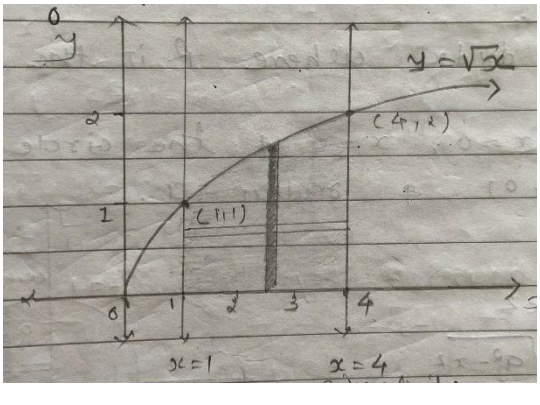
→ The point of intersection of $y^2=x$ and $x=1, x=4$

$y^2 = x \Rightarrow y=\sqrt{x}$

$x=1 \Rightarrow y=1$

$x=4 \Rightarrow y=2$

Therefore, The point of intersection are $(1,1),(4,2)$.



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$$I=\int_1^4\int_0^{\sqrt{x}}\frac{3}{2}e^{\frac{y}{\sqrt{x}}}dydx$$

$$=\int_1^4\left[\int_0^{\sqrt{x}}\frac{3}{2}e^{\frac{y}{\sqrt{x}}}dy\right]dx$$

$$=\frac{3}{2}\int_1^4\left[\frac{e^{\frac{y}{\sqrt{x}}}}{\frac{1}{\sqrt{x}}}\right]_0^{\sqrt{x}}dx$$

$$=\frac{3}{2}\int_1^4\sqrt{x}\left[e^{\frac{y}{\sqrt{x}}}\right]_0^{\sqrt{x}}dx$$

$$=\frac{3}{2}\int_1^4\sqrt{x}\left[e^{\frac{\sqrt{x}}{\sqrt{x}}}-e^0\right]dx$$

$$=\frac{3}{2}\int_1^4\sqrt{x}\left[e-1\right]dx$$

$$=\frac{3}{2}(e-1)\int_1^4 x^{\frac{1}{2}}dx$$

$$=\frac{3}{2}(e-1)\left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_1^4$$

$$=\frac{3}{2}(e-1)\frac{2}{3}\left[4^{\frac{3}{2}}-1^{\frac{3}{2}}\right]$$

$$=(e-1)\left[2^{2\cdot\frac{3}{2}}-1\right]$$

$$=(e-1)\left[2^3-1\right]$$

$$=(e-1)\left[8-1\right]$$

$$=7(e-1)$$

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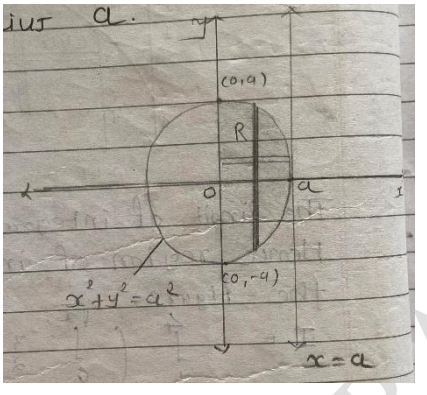
Example 5: Evaluate $\iint_R dydx$ where R is the region bounded by $x=0, x=a$ and the circle with centre $(0,0)$ and radius a.

Solution:



→ The circle with centre $(0,0)$ and radius a is $x^2+y^2=a^2$

→ $x^2+y^2=a^2 \Rightarrow y^2=a^2-x^2 \Rightarrow y=\pm\sqrt{a^2-x^2}$

→ Draw a vertical strip parallel to y-axis which starts from the part of the circle $x^2+y^2=a^2$ below x-axis and terminates on the part of the circle $x^2+y^2=a^2$ above x-axis.

→ Limit of $y: y=-\sqrt{a^2-x^2}$ to $y=\sqrt{a^2-x^2}$

→ Limit of $x: x=0$ to $x=a$

$$I=\iint_R dydx$$

$$=\int_0^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dydx$$

$$=\int_0^a \left[\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dy\right] dx$$

$$=\int_0^a [y]_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dx$$

$$=\int_0^a \left[\sqrt{a^2-x^2}+\sqrt{a^2-x^2}\right] dx$$

$$=\int_0^a 2\sqrt{a^2-x^2}\,dx$$

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$$=2\left[\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_0^a$$

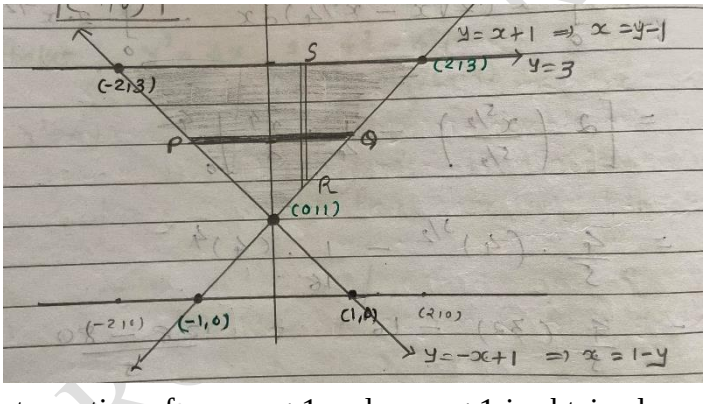
$$=2\left[\frac{a}{2}\sqrt{a^2-a^2}+\frac{a^2}{2}\sin^{-1}\frac{a}{a}-0-\frac{a^2}{2}\sin^{-1}0\right]$$

$$=2\left[\frac{a^2}{2}\sin^{-1}1-\frac{a^2}{2}(0)\right]$$

$$=2\left[\frac{a^2}{2}\left(\frac{\pi}{2}\right)\right]=\frac{a^2\pi}{2}$$

Example 6: Evaluate $\iint_R \left(2x-y^2\right)dA$; over the triangular region R enclosed between the lines $y=-x+1, y=x+1$ and $y=3$.

Solution:



→ The points of intersection of $y=-x+1$ and $y=x+1$ is obtained as

$$-x+1=x+1$$

$$\Rightarrow 2x=0$$

$$\Rightarrow x=0 \Rightarrow y=1$$

Therefore, the point of intersection is $(0,1)$.

→ The points of intersection of $y=x+1$ and $y=3$ is obtained as

$$3=x+1$$

$$\Rightarrow x=2 \Rightarrow y=3$$

Therefore, the point of intersection is $(2,3)$.

→ The points of intersection of $y=-x+1$ and $y=3$ is obtained as

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$$3 = -x + 1$$
$$\Rightarrow x = -2 \Rightarrow y = 3$$

Therefore, the point of intersection is $(-2,3)$.

→ The integration can be done w.r.t. any variable first. But in case of vertical strip, we need to divide the region into two parts. Therefore, draw a horizontal strip parallel to x-axis which starts from the line $y = -x + 1$ and terminates on the line $y = x + 1$.

→ Limits of $x : x = 1 - y$ to $x = y - 1$

→ Limits of $y : y = 1$ to $y = 3$

$$I = \iint_R \left(2x - y^2\right) dA$$

$$= \int_1^3 \int_{1-y}^{y-1} \left(2x - y^2\right) dx dy$$

$$= \int_1^3 \left[\int_{1-y}^{y-1} \left(2x - y^2\right) dx \right] dy$$

$$= \int_1^3 \left[2\frac{x^2}{2} - y^2 x \right]_{1-y}^{y-1} dy$$

$$= \int_1^3 \left\{ \left[(y-1)^2 - y^2 (y-1) \right] - \left[(1-y)^2 - y^2 (1-y) \right] \right\} dy$$

$$= \int_1^3 \left[(y-1)^2 - y^3 + y^2 - (y-1)^2 + y^2 - y^3 \right] dy$$

$$= \int_1^3 \left[-2y^3 + 2y^2 \right] dy$$

$$= \left[-2\frac{y^4}{4} + 2\frac{y^3}{3} \right]_1^3$$

$$= \left[-\frac{1}{2}3^4 + \frac{2}{3}3^3 \right] - \left[-\frac{1}{2}1^4 + \frac{2}{3}1^3 \right]$$

$$= -\frac{81}{2} + \frac{54}{3} + \frac{1}{2} - \frac{2}{3}$$

$$= -40 + \frac{52}{3} = -\frac{68}{3}$$

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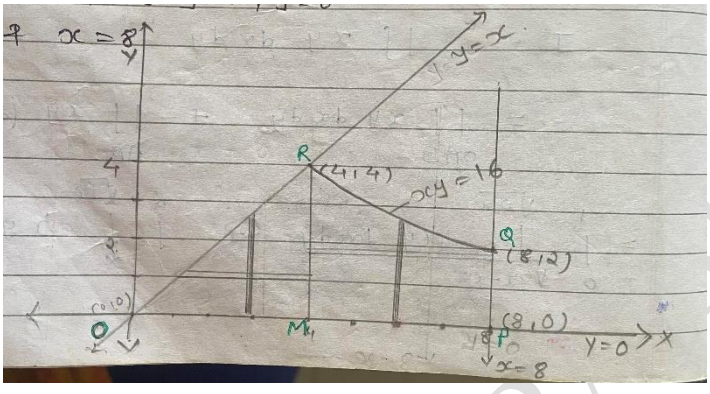
Example7: Evaluate $\iint_R x^2 dxdy$ where R is the region in the first quadrant enclosed by the rectangular hyperbola $xy = 16$, the lines $y = x, y = 0$ and $x = 8$.

Solution:



→ The region is OPQR.

→ The points of intersection of the hyperbola $xy = 16$ and the line $y = x$ are obtained as

$$xy = 16$$
$$\Rightarrow x^2 = 16$$
$$\Rightarrow x = \pm 4$$

Hence, $R(4,4)$ is the point of intersection in the first quadrant.

→ The points of intersection of the hyperbola $xy = 16$ and the line $x = 8$ are obtained as

$$xy = 16$$
$$\Rightarrow 8y = 16$$
$$\Rightarrow y = 2$$

Hence, $Q(8,2)$ is the point of intersection.

→ The integration can be done w.r.t to any variable first. To integrate w.r.t. y first we need to draw a vertical strip in the region. But here one vertical strip cannot cover the entire region, and therefore divide the region OPQR into two subregions OMR and RMPQ and draw one vertical strip in each subregion.

→ In the subregion OMR, strip starts from x-axis i.e. $y = 0$ and terminates on the line $y = x$.

Limits of $y : y = 0$ to $y = x$

Limits of $x : x = 0$ to $x = 4$

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→ In the subregion RMPQ, strip starts from x-axis i.e. $y=0$ and terminates on the Rectangular hyperbola $xy=16$.

Limits of $y: y=0$ to $y=\frac{16}{x}$

Limits of $x: x=4$ to $x=8$

$$I=\iint_R x^2 dxdy$$

$$=\iint_{OMR} x^2 dxdy+\iint_{RMPQ} x^2 dxdy$$

$$=\int_0^4\int_0^x x^2 dydx+\int_4^8\int_0^{\frac{16}{x}} x^2 dydx$$

$$=\int_0^4\left[\int_0^x x^2 dy\right]dx+\int_4^8\left[\int_0^{\frac{16}{x}} x^2 dy\right]dx$$

$$=\int_0^4 x^2\left[y\right]_0^x dx+\int_4^8 x^2\left[y\right]_0^{\frac{16}{x}} dx$$

$$=\int_0^4 x^2\left[x-0\right]dx+\int_4^8 x^2\left[\frac{16}{x}-0\right]dx$$

$$=\int_0^4 x^3 dx+\int_4^8 16x\,dx$$

$$=\left[\frac{x^4}{4}\right]_0^4+16\left[\frac{x^2}{2}\right]_4^8$$

$$=\frac{1}{4}\left[256-0\right]+8\left[64-16\right]$$

$$=\frac{1}{4}\left[256\right]+8\left[48\right]$$

$$=64+384$$

$$=448$$

Example 8: Evaluate $\iint xy\,dxdy$, over the region enclosed by the circle $x^2+y^2-2x=0$, the parabola $y^2=2x$ and the line $y=x$.

Solution:

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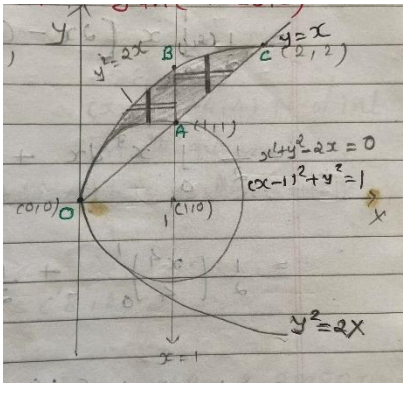

→ The points of intersection of the circle $x^2+y^2-2x=0$ and the line $y=x$ are obtained as

$$x^2+x^2-2x=0$$
$$\Rightarrow 2x^2-2x=0$$
$$\Rightarrow x(x-1)=0$$
$$\Rightarrow x=0,1 \Rightarrow y=0,1$$

Therefore, the points of intersection are $O(0,0)$ and $A(1,1)$.

→ The points of intersection of the circle $x^2+y^2-2x=0$ and the parabola $y^2=2x$ is obtained as

$$x^2+2x-2x=0$$
$$\Rightarrow x^2=0$$
$$\Rightarrow x=0 \Rightarrow y=0$$

Therefore, the point of intersection is $O(0,0)$.

→ The points of intersection of the parabola $y^2=2x$ and the line $y=x$ is obtained as

$$x^2=2x$$
$$\Rightarrow x^2-2x=0$$
$$\Rightarrow x(x-2)=0$$
$$\Rightarrow x=0,2 \Rightarrow y=0,2$$

Therefore, the points of intersection are $O(0,0)$ and $C(2,2)$.

→ The integration can be done w.r.t. any variable first. To integrate w.r.t. y first we need to draw a vertical strip in the region. But one vertical strip does not cover the entire region,

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therefore, divide the region OACBO into two subregions OAB and BAC and draw one vertical strip in each subregion.

→ In subregion OAB, strip starts from the circle $x^2+y^2-2x=0$ and terminates on the parabola $y^2=2x$.

Limits of $y: y=\sqrt{2x-x^2}$ to $y=\sqrt{2x}$

Limits of $x: x=0$ to $x=1$

→ In subregion BAC, strip starts from the line $y=x$ and terminates on the parabola $y^2=2x$.

Limits of $y: y=x$ to $y=\sqrt{2x}$

Limits of $x: x=1$ to $x=2$

$$I=\iint xy\,dxdy$$

$$=\iint_{OAB} xy\,dxdy+\iint_{BAC} xy\,dxdy$$

$$=\int_0^1\int_{\sqrt{2x-x^2}}^{\sqrt{2x}} xy\,dydx+\int_1^2\int_x^{\sqrt{2x}} xy\,dydx$$

$$=\int_0^1\left[\int_{\sqrt{2x-x^2}}^{\sqrt{2x}} xy\,dy\right]dx+\int_1^2\left[\int_x^{\sqrt{2x}} xy\,dy\right]dx$$

$$=\int_0^1 x\left[\frac{y^2}{2}\right]_{\sqrt{2x-x^2}}^{\sqrt{2x}}dx+\int_1^2 x\left[\frac{y^2}{2}\right]_x^{\sqrt{2x}}dx$$

$$=\frac{1}{2}\int_0^1 x\left[\left(\sqrt{2x}\right)^2-\left(\sqrt{2x-x^2}\right)^2\right]dx+\frac{1}{2}\int_1^2 x\left[\left(\sqrt{2x}\right)^2-(x)^2\right]dx$$

$$=\frac{1}{2}\int_0^1 x\left[2x-2x+x^2\right]dx+\frac{1}{2}\int_1^2 x\left[2x-x^2\right]dx$$

$$=\frac{1}{2}\int_0^1 x^3dx+\frac{1}{2}\int_1^2\left(2x^2-x^3\right)dx$$

$$=\frac{1}{2}\left[\frac{x^4}{4}\right]_0^1+\frac{1}{2}\left[2\frac{x^3}{3}-\frac{x^4}{4}\right]_1^2$$

$$=\frac{1}{8}[1-0]+\frac{1}{2}\left[\frac{2}{3}(8)-\frac{1}{4}(16)-\frac{2}{3}(1)+\frac{1}{4}(1)\right]$$

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$$= \frac{1}{8} + \frac{8}{3} - 2 - \frac{1}{3} + \frac{1}{8}$$

$$= \frac{7}{12}$$

H.W.

Example 9: Integrate $f(x,y) = x^2 + y^2$ over the triangular region with vertices $(0,0),(1,0)$ and $(0,1)$

Ans: $\frac{1}{6}$

Example 10: Evaluate $\iint_R x\,dxdy$, where R is bounded by $x=0, x=\frac{y^2}{4}, y=0, y=4$.

Ans: $\frac{32}{5}$

Example 11: Evaluate $\iint_R dxdy$ where R is the region bounded by $y=0, y=1$ and the circle with centre $(1,0)$ and radius 1.

Ans: $\frac{\pi}{2}$

Example 12: Evaluate $\iint_R (x^2+y^2)\,dxdy$ where R is the region bounded by the lines $y=4x, x+y=3, y=0$ and $y=2$.

Ans: $\frac{463}{48}$

Example 13: Use double integration to determine the area of the region R inclosed between the parabola $y = \frac{1}{2}x^2$, and the line $y = 2x$, also sketch the region.

Ans: $\frac{16}{3}$

Example 14: Evaluate $\iint (x^2 - y^2)\,dxdy$ over the triangle with vertices $(0,1),(1,1),(1,2)$

Ans: $-\frac{2}{3}$

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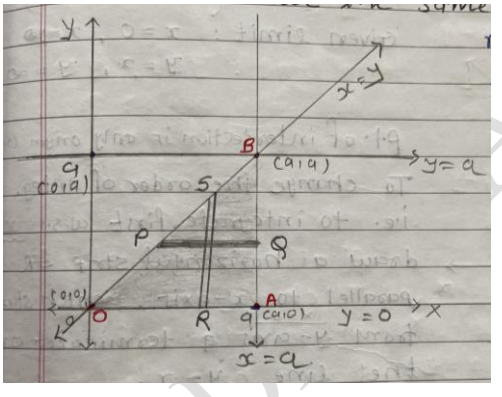
5. Change of order of integration

Sometimes, evaluation of double integral becomes easier by changing the order of integration. To change the order of integration, first, we draw the region of integration with the help of the given limits. Then we draw a vertical or horizontal strip as per the required order of integration. This change of order also changes the limits of integration.

Example 1: Change the order of integration $\int_0^a \int_y^a \frac{x}{x^2+y^2}\,dxdy$ and hence evaluate the same.

Solution:



→ Since inner limits depend on y, the function is integrated first w.r.t. x **but** evaluation becomes easier by changing the order of integration.

→ Limits of $x: x=y$ to $x=a$

Limits of $y: y=0$ to $y=a$

→ The region is bounded by the lines $y=x, x=a$ and $y=0$.

→ The point of intersection of $y=x$ and $x=a$ is $x=a=y$. Therefore, the point of intersection is $B(a,a)$.

→ To change the order of integration, i.e. to integrate first w.r.t. y, draw a vertical strip RS parallel to y-axis which starts from x-axis i.e. $y=0$ and terminates on the line $y=x$.

→ Limit of $y: y=0$ to $y=x$

→ Limit of $x: x=0$ to $x=a$

Hence, the given integral after change of order is

$$\int_0^a \int_y^a \frac{x}{x^2+y^2}\,dxdy = \int_0^a \int_0^x \frac{x}{x^2+y^2}\,dydx$$

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$$= \int_0^a x \left[\int_0^x \frac{1}{x^2 + y^2} dy \right] dx$$

$$= \int_0^a x \left[\frac{1}{x} \tan^{-1} \frac{y}{x} \right]_0^x dx$$

$$= \int_0^a \left[\tan^{-1} 1 - \tan^{-1} 0 \right] dx$$

$$= \int_0^a \left[\frac{\pi}{4} - 0 \right] dx$$

$$= \frac{\pi}{4} \int_0^a dx$$

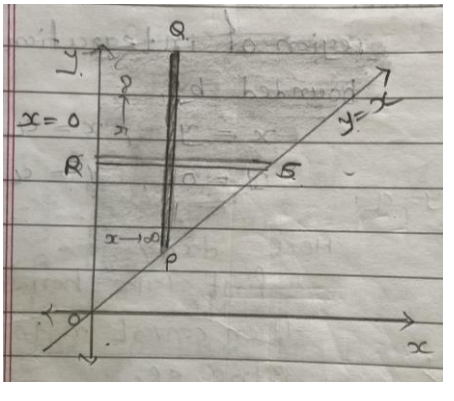
$$= \frac{\pi}{4} [x]_0^a$$

$$= \frac{\pi}{4} [a - 0]$$

$$= \frac{\pi a}{4}$$

Example 2: Change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dydx$.

Solution:



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→ Since the inner limits depend on x, the function is integrated first w.r.t. y **but** evaluation becomes easier by changing the order of integration.

→ Limits of $y: y = x$ to $y \to \infty$

→ Limits of $x: x = 0$ to $x \to \infty$

→ The region is bounded by the lines $y = x$ and $x = 0$.

→ Here, the only point of intersection is origin.

→ To change the order of integration, i.e. to integrate first w.r.t. x, draw a horizontal strip RS parallel to x-axis which starts from y-axis i.e. $x = 0$ and terminates on the line $y = x$.

→ Limits of $x: x = 0$ to $x = y$

→ Limits of $y: y = 0$ to $x \to \infty$

Hence, the given integral after change of order is

$$\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dydx$$

$$= \int_0^\infty \int_0^y \frac{e^{-y}}{y} dxdy$$

$$= \int_0^\infty \frac{e^{-y}}{y} \left[\int_0^y dx \right] dy$$

$$= \int_0^\infty \frac{e^{-y}}{y} [x]_0^y \, dy$$

$$= \int_0^\infty \frac{e^{-y}}{y} [y - 0] dy$$

$$= \int_0^\infty e^{-y} dy$$

$$= \left[-e^{-y} \right]_0^\infty$$

$$= -\left[e^{-\infty} - e^0 \right]$$

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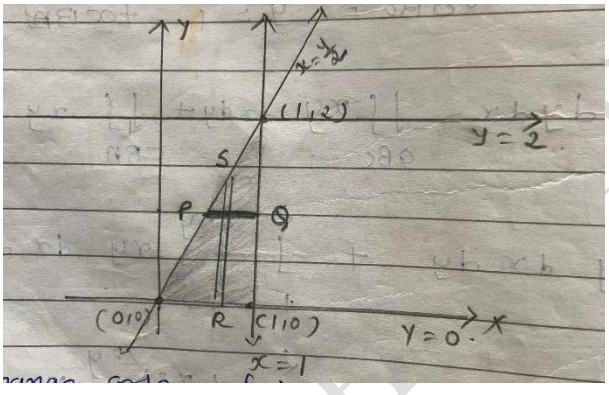
$= -[0-1]$

$= 1$

Example 3: Evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2}\, dxdy$ by changing the order of integration.

Solution:



→ The function is integrated first w.r.t. x, but evaluation becomes easier by changing the order of integration.

→ Limits of $x : x = \frac{y}{2}$ to $x = 1$

→ Limits of $y : y = 0$ to $y = 2$

→ The region is bounded by the lines $y = 2x, x = 1, y = 2$ and $y = 0$.

→ The point of intersection of $y = 2x$ and $x = 1$ is $x = 1, y = 2$. Therefore, the point of intersection is $(1, 2)$.

→ To change the order of intersection, i.e. to integrate first w.r.t. y, draw a vertical strip RS parallel to y-axis which starts from x-axis i.e. $y = 0$ and terminates on the line $y = 2x$.

→ Limits of $y : y = 0$ to $y = 2x$

→ Limits of $x : x = 0$ to $x = 1$

Hence, the given integral after change of order is

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$$\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2}\,dxdy$$

$$= \int_0^1 \int_0^{2x} e^{x^2}\,dydx$$

$$= \int_0^1 \left[\int_0^{2x} e^{x^2}\,dy \right] dx$$

$$= \int_0^1 e^{x^2} \left[y \right]_0^{2x} dx$$

$$= \int_0^1 e^{x^2} \left[2x - 0 \right] dx$$

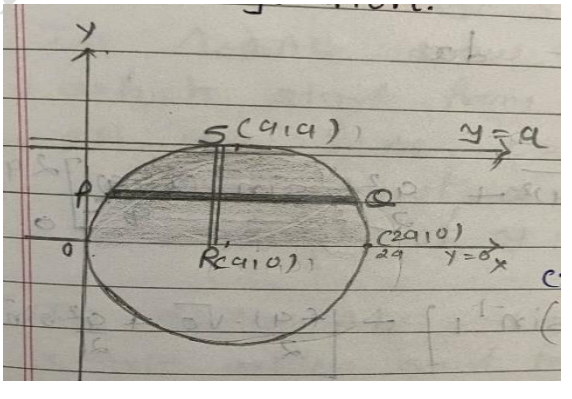
$$= \left[e^{x^2} \right]_0^1 \qquad \left(\because \int e^{f(x)} f'(x)\,dx = e^{f(x)} \right)$$

$$= e^1 - e^0$$

$$= e - 1$$

Example 4: Evaluate $\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} dydx$ by changing the order of integration.

Solution:



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→ Since inner limits depend on y, the function is integrated first w.r.t. x.

The correct form of the integral is $\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} dxdy$

→ Limits of $x: x = a-\sqrt{a^2-y^2}$ to $x = a+\sqrt{a^2-y^2}$

→ Limits of $y: y = 0$ to $y = a$

→ The region is bounded by the circle $(x-a)^2 + y^2 = a^2$ and the line $y = 0$. Since limits of y are positive, the region is the part of the circle $(x-a)^2 + y^2 = a^2$ above x-axis.

→ The points of intersection of the circle with x-axis are $(0,0)$ and $(2a,0)$.

→ To change the order of intersection i.e. integrate first w.r.t. y, draw a vertical strip RS parallel to y-axis which starts from x-axis i.e. $y = 0$ and terminates on circle

$$(x-a)^2 + y^2 = a^2$$

$$\Rightarrow y^2 = a^2 - (x-a)^2$$

$$\Rightarrow y^2 = a^2 - x^2 + 2ax - a^2$$

$$\Rightarrow y^2 = 2ax - x^2$$

$$\Rightarrow y = \sqrt{2ax - x^2}$$

→ Limits of $y: y = 0$ to $y = \sqrt{2ax-x^2}$

→ Limits of $x: x = 0$ to $x = 2a$

Hence, the given integral after change of order is

$$\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} dx\, dy$$

$$= \int_0^{2a} \int_0^{\sqrt{2ax-x^2}} dydx$$

$$= \int_0^{2a} \left[\int_0^{\sqrt{2ax-x^2}} dy \right] dx$$

$$= \int_0^{2a} [y]_0^{\sqrt{2ax-x^2}} dx$$

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$$= \int_0^{2a} \left[\sqrt{2ax - x^2} - 0 \right] dx$$

$$= \int_0^{2a} \sqrt{a^2 - a^2 + 2ax - x^2}\, dx$$

$$= \int_0^{2a} \sqrt{a^2 - (x-a)^2}\, dx$$

$$= \left[\frac{x-a}{2}\sqrt{a^2-(x-a)^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x-a}{a}\right) \right]_0^{2a}$$

$$= \left[\frac{2a-a}{2}\sqrt{a^2-(2a-a)^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{2a-a}{a}\right) \right] - \left[\frac{0-a}{2}\sqrt{a^2-(0-a)^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{0-a}{a}\right) \right]$$

$$= \left[\frac{a}{2}\sqrt{0} + \frac{a^2}{2}\sin^{-1}1 \right] - \left[-\frac{a}{2}\sqrt{0} + \frac{a^2}{2}\sin^{-1}(-1) \right]$$

$$= \left[\frac{a^2}{2}\sin^{-1}1 \right] - \left[-\frac{a^2}{2}\sin^{-1}1 \right]$$

$$= 2\left(\frac{a^2}{2} sin^{-1} 1 \right)$$

$$= \frac{\pi a^2}{2}$$

Example 5: Evaluate $\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dydx$ by changing the order of integration.

Solution:

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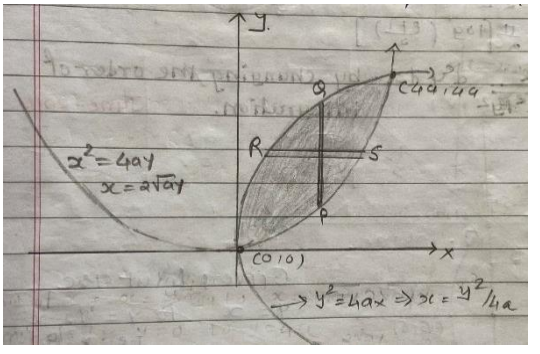

→ Since inner limits depend on x, the function is integrated first w.r.t. y.

→ Limits of $y : y = \dfrac{x^2}{4a}$ to $y = 2\sqrt{ax}$

→ Limits of $x : x = 0$ to $x = 4a$

→ The region is bounded by the parabolas $x^2 = 4ay$ and $y^2 = 4ax$.

→ The points of intersection of $x^2 = 4ay$ and $y^2 = 4ax$ are obtained as

$(x^2)^2 = (4ay)^2 \Longrightarrow x^4 = 16a^2y^2$

$\Rightarrow x^4 = 16a^2(4ax)$

$\Rightarrow x^4 - 64a^3x = 0$

$\Rightarrow x(x^3 - 64a^3) = 0$

$\Rightarrow x = 0, x = 4a \Rightarrow y = 0, 4a$

The points of intersection are $(0,0)$ and $(4a, 4a)$.

→ To change the order of integration i.e. to integrate first w.r.t. x, draw a horizontal strip RS parallel to x-axis which starts from the parabola $y^2 = 4ax$ and terminates on the parabola $x^2 = 4ay$.

→ Limits of $x : x = \dfrac{y^2}{4a}$ to $x = 2\sqrt{ay}$

→ Limits of $y : y = 0$ to $y = 4a$

Hence, the given integral after change of order is

$$\int_0^{4a} \int_{\frac{x^2}{4a}}^{2\sqrt{ax}} dydx$$

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$$= \int_0^{4a} \int_{\frac{y^2}{4a}}^{2\sqrt{ay}} dxdy$$

$$= \int_0^{4a} \left[\int_{\frac{y^2}{4a}}^{2\sqrt{ay}} dx \right] dy$$

$$= \int_0^{4a} [x]_{\frac{y^2}{4a}}^{2\sqrt{ay}} dy$$

$$= \int_0^{4a} \left[2\sqrt{ay} - \frac{y^2}{4a} \right] dy$$

$$= \left[2\sqrt{a} \frac{y^{\frac{3}{2}}}{\frac{3}{2}} - \frac{1}{4a} \frac{y^3}{3} \right]_0^{4a}$$

$$= \left[\frac{4}{3}\sqrt{a}(4a)^{\frac{3}{2}} - \frac{1}{12a}(4a)^3 \right] - [0]$$

$$= \frac{32}{3}a^2 - \frac{16}{3}a^2$$

$$= \frac{16}{3}a^2$$

Example 6: Change the order of integration in the integral $\int_0^1 \int_{x^2}^{2-x} xy\, dxdy$ and then evaluate it.

Solution:

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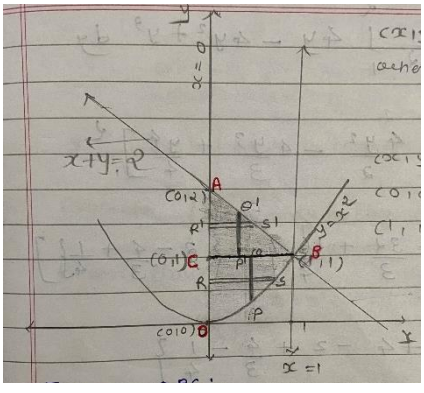

→ Since inner limits depend on x, the function is integrated first w.r.t. y.

The correct form of integral is $\int_0^1 \int_{x^2}^{2-x} xy\,dydx$.

→ Limits of $y: y = x^2$ to $y = 2 - x$

→ Limits of $x: x = 0$ to $x = 1$

→ The region is bounded by y-axis, the line $x + y = 2$ and the parabola $x^2 = y$. Since given limits of x and y are positive, the region lies in the first quadrant.

→ The points of intersection of $x + y = 2$ and $x^2 = y$ are obtained as

$$x^2 = 2 - x$$
$$\Rightarrow x^2 + x - 2 = 0$$
$$\Rightarrow (x-1)(x+2) = 0$$
$$\Rightarrow x = 1, -2 \Rightarrow y = 1, 4$$

Therefore, the points of intersection are $B(1,1)$ and $(-2,4)$.

→ To change the order of intersection, i.e. to integrate first w.r.t. x, divide the region into two subregion OBC and CBA. Draw a horizontal strip parallel to x-axis in each subregion.

→ In subregion OBC, strip RS starts from y-axis i.e. $x = 0$ and terminates on parabola $x^2 = y$.

Limits of $x: x = 0$ to $x = \sqrt{y}$

Limits of $y: y = 0$ to $y = 1$

→ In subregion CBA, strip R'S' starts from y-axis i.e. $x = 0$ and terminates on the line $x + y = 2$.

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Limit of $x : x = 0$ to $x = 2 - y$

Limits of $y : y = 1$ to $y = 2$

Hence, the given integral after change of order is

$$\int_0^1 \int_{x^2}^{2-x} xy\, dydx$$

$$= \int_0^1 \int_0^{\sqrt{y}} xy\, dxdy + \int_1^2 \int_0^{2-y} xy\, dxdy$$

$$= \int_0^1 \left[\int_0^{\sqrt{y}} xy\, dx \right] dy + \int_1^2 \left[\int_0^{2-y} xy\, dx \right] dy$$

$$= \int_0^1 y \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} dy + \int_1^2 y \left[\frac{x^2}{2} \right]_0^{2-y} dy$$

$$= \frac{1}{2} \int_0^1 y[y - 0]\, dy + \frac{1}{2} \int_1^2 y\left[(2-y)^2 - 0 \right] dy$$

$$= \frac{1}{2} \int_0^1 y^2 dy + \frac{1}{2} \int_1^2 y\left(4 - 4y + y^2\right) dy$$

$$= \frac{1}{2} \int_0^1 y^2 dy + \frac{1}{2} \int_1^2 \left(4y - 4y^2 + y^3\right) dy$$

$$= \frac{1}{2} \left[\frac{y^3}{3} \right]_0^1 + \frac{1}{2} \left[4\frac{y^2}{2} - 4\frac{y^3}{3} + \frac{y^4}{4} \right]_1^2$$

$$= \frac{1}{6}(1 - 0) + \frac{1}{2} \left[2(4-1) - \frac{4}{3}(8-1) + \frac{1}{4}(16-1) \right]$$

$$= \frac{1}{6}(1) + \frac{1}{2} \left[2(3) - \frac{4}{3}(7) + \frac{1}{4}(15) \right]$$

$$= \frac{1}{6} + \frac{1}{2} \left[\frac{72 - 112 + 45}{12} \right]$$

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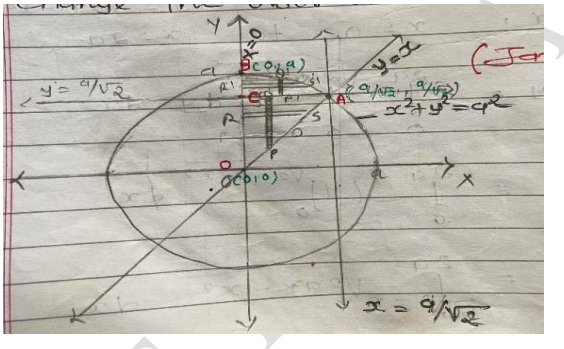
$$= \frac{1}{6} + \frac{5}{24}$$

$$= \frac{9}{24}$$

$$= \frac{3}{8}$$

Example 7: Change the order of integral and evaluate $\int_0^{\frac{a}{\sqrt{2}}} \int_x^{\sqrt{a^2-x^2}} y^2 dA$.

Solution:



→ Since inner limits depend on x, the function is integrated first w.r.t. y.

The correct form of integral is $\int_0^{\frac{a}{\sqrt{2}}} \int_x^{\sqrt{a^2-x^2}} y^2 dydx$.

→ Limits of $y : y = x$ to $y = \sqrt{a^2 - x^2}$

→ Limits of $x : x = 0$ to $x = \frac{a}{\sqrt{2}}$

→ The region is bounded by y-axis, the circle $x^2 + y^2 = a^2$, the lines $y = x, x = \frac{a}{\sqrt{2}}$. Since given limits of x and y are positive, the region lies in the first quadrant.

→ The points of intersection of the circle $x^2 + y^2 = a^2$, the line $y = x$ are obtained as

$$2x^2 = a^2$$

$$\Rightarrow x^2 = \frac{a^2}{2}$$

$\Rightarrow x = \pm \frac{a}{\sqrt{2}}$

Therefore, the points of intersection are $A\left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$ and $\left(-\frac{a}{\sqrt{2}}, -\frac{a}{\sqrt{2}}\right)$.

→ To change the order of intersection, i.e. to integrate first w.r.t. x, divide the region into two subregion OAC and ABC. Draw a horizontal strip parallel to x-axis in each subregion.

→ In subregion OAC, strip RS starts from y-axis i.e. $x = 0$ and terminates on the line $y = x$.

Limits of $x : x = 0$ to $x = y$

Limits of $y : y = 0$ to $y = \frac{a}{\sqrt{2}}$

→ In subregion ABC, strip R'S' starts from y-axis i.e. $x = 0$ and terminates on the circle $x^2 + y^2 = a^2$.

Limits of $x : x = 0$ to $x = \sqrt{a^2 - y^2}$

Limits of $y : y = \frac{a}{\sqrt{2}}$ to $y = a$

Hence, the given integral after change of order is

$$\int_0^{\frac{a}{\sqrt{2}}} \int_x^{\sqrt{a^2-x^2}} y^2 dydx$$

$$= \int_0^{\frac{a}{\sqrt{2}}} \int_0^{y} y^2 dxdy + \int_{\frac{a}{\sqrt{2}}}^{a} \int_0^{\sqrt{a^2-y^2}} y^2 dxdy$$

$$= \int_0^{\frac{a}{\sqrt{2}}} \left[\int_0^{y} y^2 dx \right] dy + \int_{\frac{a}{\sqrt{2}}}^{a} \left[\int_0^{\sqrt{a^2-y^2}} y^2 dx \right] dy$$

$$= \int_0^{\frac{a}{\sqrt{2}}} y^2 [x]_0^y dy + \int_{\frac{a}{\sqrt{2}}}^{a} y^2 [x]_0^{\sqrt{a^2-y^2}} dy$$

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$$= \int_0^{\frac{a}{\sqrt{2}}} y^2 [y-0]\,dy + \int_{\frac{a}{\sqrt{2}}}^{a} y^2 \left[\sqrt{a^2-y^2}-0\right] dy$$

$$= \int_0^{\frac{a}{\sqrt{2}}} y^3 dy + \int_{\frac{a}{\sqrt{2}}}^{a} y^2 \sqrt{a^2-y^2}\,dy$$

In second term

Put $y = a\sin\theta$

$\Rightarrow dy = a\cos\theta d\theta$

When $y = \dfrac{a}{\sqrt{2}}$ then $\theta = \dfrac{\pi}{4}$ and when $y = a$ then $\theta = \dfrac{\pi}{2}$

$$= \int_0^{\frac{a}{\sqrt{2}}} y^3 dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (a\sin\theta)^2 \sqrt{a^2-(a\sin\theta)^2}\,(a\cos\theta\, d\theta)$$

$$= \int_0^{\frac{a}{\sqrt{2}}} y^3 dy + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} a^2\sin^2\theta \cdot a\sqrt{1-\sin^2\theta}\,(a\cos\theta\, d\theta)$$

$$= \int_0^{\frac{a}{\sqrt{2}}} y^3 dy + a^4 \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2\theta \cdot \cos\theta \cdot \cos\theta\, d\theta$$

$$= \int_0^{\frac{a}{\sqrt{2}}} y^3 dy + \frac{a^4}{4} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 4\sin^2\theta\cos^2\theta\, d\theta$$

$$= \int_0^{\frac{a}{\sqrt{2}}} y^3 dy + \frac{a^4}{4} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 2\theta d\theta \qquad (\because 2\sin\theta\cos\theta = \sin 2\theta)$$

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$$= \int_0^{\frac{a}{\sqrt{2}}} y^3 dy + \frac{a^4}{4}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1-\cos 4\theta}{2} d\theta \quad \left(\because \sin^2\theta = \frac{1-\cos 2\theta}{2}\right)$$

$$= \left[\frac{y^4}{4}\right]_0^{\frac{a}{\sqrt{2}}} + \frac{a^4}{8}\left[\theta - \frac{\sin 4\theta}{4}\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \frac{1}{4}\left[\frac{a^4}{4} - 0\right] + \frac{a^4}{8}\left[\left(\frac{\pi}{2} - \frac{\sin 2\pi}{4}\right) - \left(\frac{\pi}{4} - \frac{\sin \pi}{4}\right)\right]$$

$$= \frac{a^4}{16} + \frac{a^4}{8}\left[\frac{\pi}{2} - 0 - \frac{\pi}{4} + 0\right]$$

$$= \frac{a^4}{16} + \frac{a^4}{8}\left[\frac{\pi}{4}\right]$$

$$= \frac{a^4}{16}\left(1 + \frac{\pi}{2}\right)$$

H.W.

Example 8: Change the order of integration $\int_0^\infty \int_x^\infty e^{-y^2} dydx$ and evaluate it.

Ans: $\frac{1}{2}$

Example 9: Evaluate the integral $\int_{y=0}^{1} \int_{x=0}^{\cos^{-1} y} e^{sinx}\ dxdy$ by change of order

Ans: $e - 1$

Example 10: Change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}} dydx$.

Ans: $\frac{\pi}{2}\log\left(\frac{e+1}{2}\right)$

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Example 11: Change the order of integration and evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{\cos^{-1} x}{\sqrt{1-x^2}\sqrt{1-x^2-y^2}} dxdy$ ..

Ans: $\frac{\pi^3}{16}$

Example 12: Evaluate $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2+y^2}} dydx$ by changing the order of integration.

Ans: $\frac{\sqrt{2}-1}{\sqrt{2}}$

6. Change of variables in double integral

➢ Change of variables in double integral by Jacobians: $(x, y \to r, \theta)$

The double integral can be changed from cartesian coordinates (x, y) to polar coordinates (r, θ) by putting $x = r\cos\theta, y = r\sin\theta$. Then

$\iint f(x,y)\,dydx = \iint f(r\cos\theta, r\sin\theta)|J|\,drd\theta$ where J is the Jacobian (functional determinant) defined as

$$J = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r$$

Hence,

$$\iint f(x,y)\,dydx = \iint f(r\cos\theta, r\sin\theta)|r|\,drd\theta = \iint f(r\cos\theta, r\sin\theta)\,r\,drd\theta$$

Example 1: Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dxdy$ by changing to polar co-ordinates.

Solution: Here, given limit: $0 \le x \le \infty, 0 \le y \le \infty$

Take $x = r\cos\theta, y = r\sin\theta \Rightarrow dxdy = r\,drd\theta$ and $x^2 + y^2 = r^2$ (1)

To find the limit of r and θ

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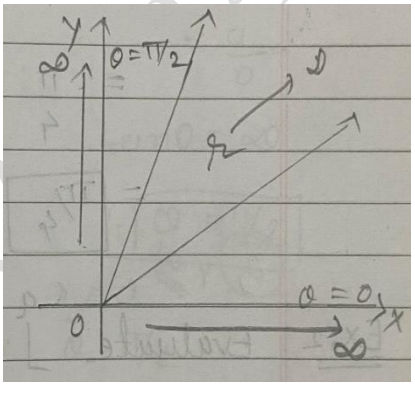
Take $x=0, y=0$ in (1), we have $r=0$

And $x=\infty, y=\infty$ in (1), we have $r=\infty$

Therefore, $0 \le r \le \infty$

Now, $\frac{x}{r}=\cos\theta, \frac{y}{r}=\sin\theta$

$$\therefore \frac{\sin\theta}{\cos\theta}=\frac{y}{x}=\tan\theta \Rightarrow \theta=\tan^{-1}\left(\frac{y}{x}\right)$$



If $x=y=0 \Rightarrow \theta=\tan^{-1}0=0$

And $x=y=\infty \Rightarrow \theta=\tan^{-1}\infty=\frac{\pi}{2}$

Therefore, $0 \le \theta \le \frac{\pi}{2}$

Hence, the given integral after transformation is

$$\int_0^\infty\int_0^\infty e^{-\left(x^2+y^2\right)}dxdy$$

$$=\int_0^{\frac{\pi}{2}}\int_0^\infty e^{-r^2}r\,drd\theta$$

$$=\frac{1}{2}\left(\int_0^\infty e^{-r^2}(2r)dr\right)\left(\int_0^{\frac{\pi}{2}}d\theta\right)$$

Take $r^2=t \Rightarrow 2r\,dr=dt$

When $r=0 \Rightarrow t=0$ and $r=\infty \Rightarrow t=\infty$

$$=\frac{1}{2}\left(\int_0^\infty e^{-t}dt\right)(\theta)_0^{\frac{\pi}{2}}$$

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$$=\frac{1}{2}\left(\frac{e^{-t}}{-1}\right)_0^{\infty}(\theta)_0^{\frac{\pi}{2}}$$

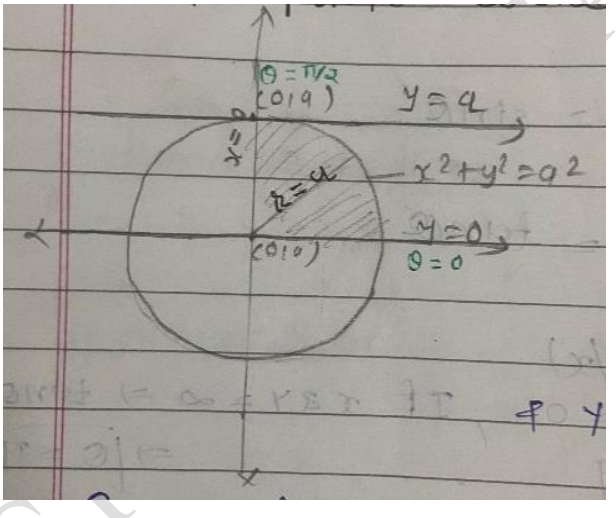
$$=-\frac{1}{2}\left(e^{-\infty}-e^{0}\right)\left(\frac{\pi}{2}-0\right)$$

$$=-\frac{\pi}{4}(0-1)=\frac{\pi}{4}$$

Example 2: Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2\sqrt{x^2+y^2}\,dxdy$ by changing in to polar co-ordinates.

Solution: Here, given limit: $x=0$ to $x=\sqrt{a^2-y^2}$ and $y=0$ to $y=a$

$x=\sqrt{a^2-y^2} \Rightarrow x^2+y^2=a^2$ (Circle with centre (0,0) and radius $r=a$)



Take $x=r\cos\theta, y=r\sin\theta \Rightarrow dxdy=r\,drd\theta$ and $x^2+y^2=r^2$ (1)

To find the limit of r and θ

Take $x=0, y=0$ in (1), we have $r=0$

And $x=\sqrt{a^2-y^2}, y=a$ in (1), we have $r=a$

Therefore, $0 \le r \le a$

Now, $\frac{x}{r} = \cos\theta, \frac{y}{r} = \sin\theta$

$$\therefore \frac{\sin\theta}{\cos\theta} = \frac{y}{x} = \tan\theta \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

If $x = y = 0 \Rightarrow \theta = \tan^{-1} 0 = 0$

And $x = \sqrt{a^2 - y^2}, y = a \Rightarrow \theta = \tan^{-1}\frac{a}{\sqrt{a^2 - a^2}} \Rightarrow \theta = \tan^{-1}\infty = \frac{\pi}{2}$

Therefore, $0 \le \theta \le \infty$

Hence, the given integral after transformation is

$$\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2 + y^2}\, dxdy$$

$$= \int_0^{\frac{\pi}{2}} \int_0^a r^2 \sin^2\theta (r)(r\, drd\theta)$$

$$= \int_0^{\frac{\pi}{2}} \left[\int_0^a r^4 \sin^2\theta\, dr\right] d\theta$$

$$= \int_0^{\frac{\pi}{2}} \sin^2\theta \left[\frac{r^5}{5}\right]_0^a d\theta$$

$$= \frac{1}{5}\int_0^{\frac{\pi}{2}} \sin^2\theta \left[a^5 - 0\right] d\theta$$

$$= \frac{a^5}{5}\int_0^{\frac{\pi}{2}} \sin^2\theta d\theta$$

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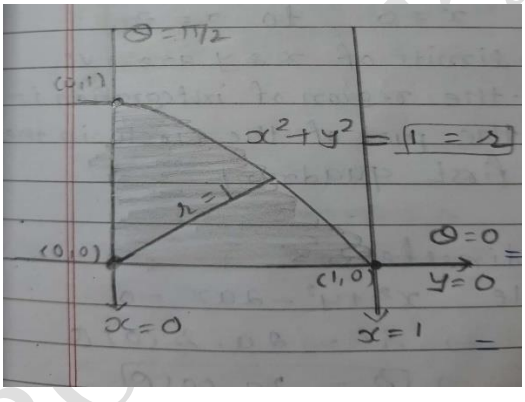
$$= \frac{a^5}{5}\left(\frac{1}{2}\cdot\frac{\pi}{2}\right)$$

$$= \frac{\pi a^5}{20}$$

Example 3: Change the cartesian integral $\int_0^1 \int_0^{\sqrt{1-x^2}} e^{-\left(x^2+y^2\right)} dydx$ in to an equivalent polar integral. Then evaluate the same.

Solution: Here, given limit: $y=0$ to $y=\sqrt{1-x^2}$ and $x=0$ to $x=1$

$y=\sqrt{1-x^2} \Rightarrow x^2+y^2=1$ (Circle with centre (0,0) and radius $r=1$)



Changing in to polar coordinates

$\rightarrow$ Limits of $r: r=0$ to $r=1$

$\rightarrow$ Limits of $\theta: \theta=0$ to $\theta=\frac{\pi}{2}$

Hence, the given integral after transformation is

$$\int_0^1 \int_0^{\sqrt{1-x^2}} e^{-\left(x^2+y^2\right)} dydx$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 e^{-r^2} \cdot r\, drd\theta$$

$$=\left(-\frac{1}{2}\int_0^1 e^{-r^2}\cdot(-2r)\,dr\right)\left(\int_0^{\frac{\pi}{2}} d\theta\right)$$

$$=\left(-\frac{1}{2}e^{-r^2}\right)_0^1 (\theta)_0^{\frac{\pi}{2}} \qquad \left(\because \int e^{f(x)} f'(x)\,dx = e^{f(x)}\right)$$

$$=-\frac{1}{2}\left(e^{-1}-e^{0}\right)\left(\frac{\pi}{2}-0\right)$$

$$=-\frac{\pi}{4}\left(\frac{1}{e}-1\right)$$

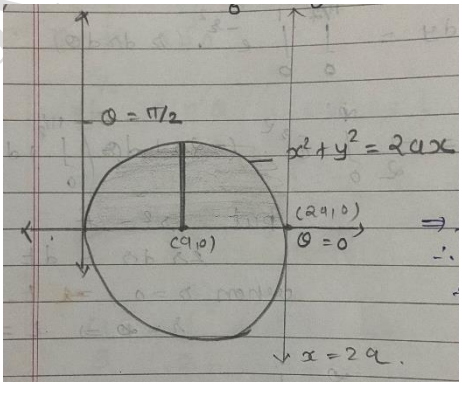
$$=\frac{\pi}{4}\left(1-\frac{1}{e}\right)$$

Example 4: By changing in to polar co-ordinates, evaluate the integral

$$\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} \left(x^2+y^2\right)dxdy\,.$$

Solution: Here, given limit: $y=0$ to $y=\sqrt{2ax-x^2}$ and $x=0$ to $x=2a$

$y=\sqrt{2ax-x^2} \Rightarrow x^2+y^2=2ax$ (Circle with centre $(a,0)$ and radius $r=a$)



Here, limits are positive. Therefore, the region of integration is the part of the circle in the first quadrant.

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Changing in to polar coordinates

→ Polar form of the circle

$x^2 + y^2 - 2ax = 0$

$\Rightarrow r^2 = 2a(r\cos\theta)$

$\Rightarrow r = 2a\cos\theta$

→ Limits of $r : r = 0$ to $r = 2a\cos\theta$

→ Limits of $\theta : \theta = 0$ to $\theta = \frac{\pi}{2}$

Hence, the given integral after transformation is

$$\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} \left(x^2+y^2\right)dxdy$$

$$= \int_0^{\frac{\pi}{2}}\int_0^{2a\cos\theta} r^2\left(r\,drd\theta\right)$$

$$= \int_0^{\frac{\pi}{2}}\left[\int_0^{2a\cos\theta} r^3dr\right]d\theta$$

$$= \int_0^{\frac{\pi}{2}}\left[\frac{r^4}{4}\right]_0^{2a\cos\theta} d\theta$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}}\left[\left(2a\cos\theta\right)^4 - 0\right]d\theta$$

$$= 4a^4\int_0^{\frac{\pi}{2}}\cos^4\theta d\theta$$

$$= 4a^4\,\frac{3\cdot 1}{4\cdot 2}\,\frac{\pi}{2} = \frac{3\pi a^4}{4}$$

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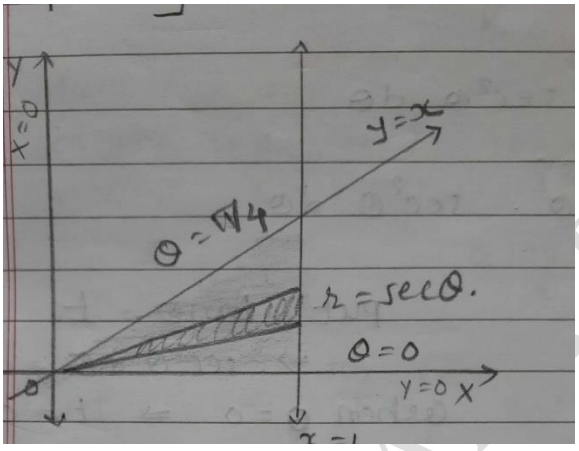
Example 5: Evaluate the integral $\int_0^1 \int_0^x \sqrt{x^2+y^2}\,dydx$ by passing on to the polar co-ordinates.

Solution: Here, given limit: $y=0$ to $y=x$ and $x=0$ to $x=1$



Now,

$x=0$
$\Rightarrow r\cos\theta=0$
$\Rightarrow r=0$

$x=1$
$\Rightarrow r\cos\theta=1$
$\Rightarrow r=\dfrac{1}{\cos\theta}=\sec\theta$

And

$y=0$
$\Rightarrow r\sin\theta=0$
$\Rightarrow \theta=0$

$y=x$
$\Rightarrow r\sin\theta=r\cos\theta$
$\Rightarrow \tan\theta=1$
$\Rightarrow \theta=\dfrac{\pi}{4}$

Changing in to polar coordinates

$\rightarrow$ Limits of $r: r=0$ to $r=\sec\theta$

$\rightarrow$ Limits of $\theta: \theta=0$ to $\theta=\dfrac{\pi}{4}$

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Hence, the given integral after transformation is

$$\int_0^1 \int_0^x \sqrt{x^2+y^2}\,dydx$$

$$=\int_0^{\frac{\pi}{4}} \int_0^{\sec\theta} \sqrt{r^2}\left(r\,drd\theta\right)$$

$$=\int_0^{\frac{\pi}{4}} \left[\int_0^{\sec\theta} r^2 dr\right] d\theta$$

$$=\int_0^{\frac{\pi}{4}} \left[\frac{r^3}{3}\right]_0^{\sec\theta} d\theta$$

$$=\frac{1}{3}\int_0^{\frac{\pi}{4}} \left(\sec^3\theta - 0\right) d\theta$$

$$=\frac{1}{3}\int_0^{\frac{\pi}{4}} \sec\theta \cdot \sec^2\theta d\theta$$

$$=\frac{1}{3}\int_0^{\frac{\pi}{4}} \sqrt{1+\tan^2\theta} \cdot \sec^2\theta d\theta$$

Put $\tan\theta = t \Rightarrow \sec^2\theta d\theta = dt$

When $\theta = 0 \Rightarrow t = 0$ and $\theta = \frac{\pi}{4} \Rightarrow t = 1$

$$=\frac{1}{3}\int_0^1 \sqrt{1+t^2}\,dt$$

$$=\frac{1}{3}\left[\frac{t}{2}\sqrt{1+t^2} + \frac{1}{2}\log\left(t+\sqrt{1+t^2}\right)\right]_0^1$$

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$$=\frac{1}{3}\left[\left(\frac{1}{2}\sqrt{1+1}+\frac{1}{2}\log\left(1+\sqrt{1+1}\right)\right)-\left(\frac{0}{2}\sqrt{1+0}+\frac{1}{2}\log\left(0+\sqrt{1+0}\right)\right)\right]$$

$$=\frac{1}{3}\left[\left(\frac{1}{2}\sqrt{2}+\frac{1}{2}\log\left(1+\sqrt{2}\right)\right)-\frac{1}{2}\log 1\right]$$

$$=\frac{1}{3}\left[\sqrt{2}+\log\left(1+\sqrt{2}\right)\right]$$

H.W.

Example 6: Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} \left(x^2+y^2\right) dxdy$ by changing in to polar co-ordinates.

Ans: $\frac{\pi a^4}{8}$

Example 7: Evaluate $\int_0^a \int_y^a \frac{x}{x^2+y^2} dydx$ by changing in to polar co-ordinates.

Ans: $\frac{\pi a}{4}$

7. Area by double integrals

➢ Area in cartesian coordinates:

(1) The area A bounded by the curves $y=y_1(x)$ and $y=y_2(x)$ intersecting at the points $P(a,b)$ and $Q(c,d)$ is

$$A=\int_a^c \int_{y_1(x)}^{y_2(x)} dydx$$

(2) If equations of the curves are represented as $x=x_1(y)$ and $x=x_2(y)$ intersecting at the points $P(a,b)$ and $Q(c,d)$ then

$$A=\int_b^d \int_{x_1(y)}^{x_2(y)} dxdy$$

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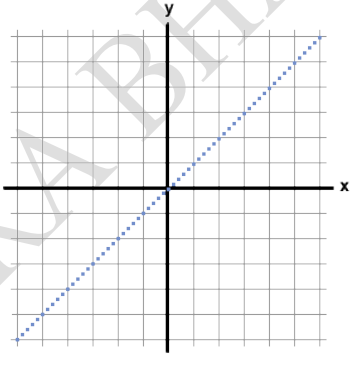
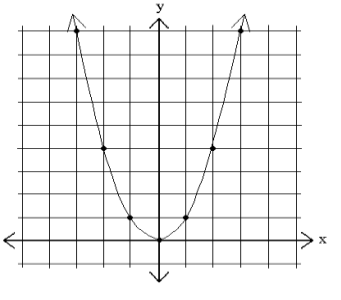
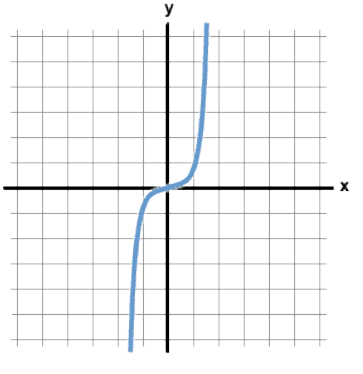
➢ Area in polar coordinates:

The area A bounded by the curves $r = r_1(\theta), r = r_2(\theta)$ and the lines $\theta = \theta_1$ and $\theta = \theta_2$ is

$$A = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} r\, dr d\theta$$

Some graph

Sr. No.	Equation	Graph
1	Straight line $y = x$	
2	Parabola $y = x^2$	
3	Cubical parabola $y = x^3$	

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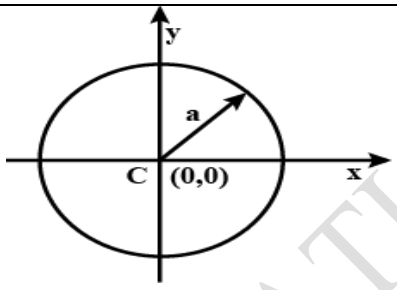
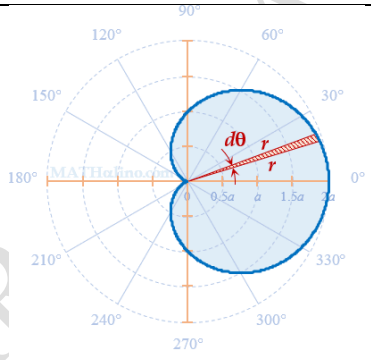
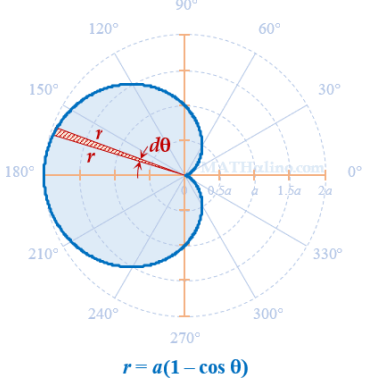
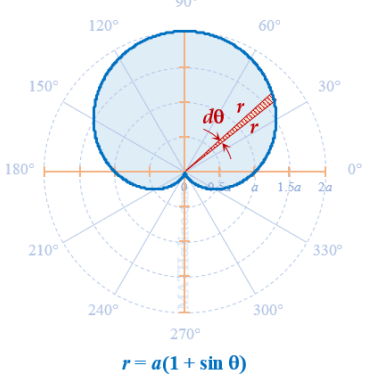
4	Circle $x^2+y^2=a^2$	
5	Cardioid $r=a(1+\cos\theta)$ $0\le\theta\le\pi$ $0\le r\le 2a$	
6	Cardioid $r=a(1-\cos\theta)$ $0\le\theta\le\pi$ $0\le r\le 2a$	
7	Cardioid $r=a(1+\sin\theta)$ $0\le\theta\le\pi$ $0\le r\le 2a$	

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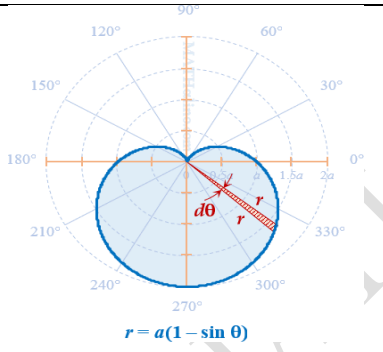
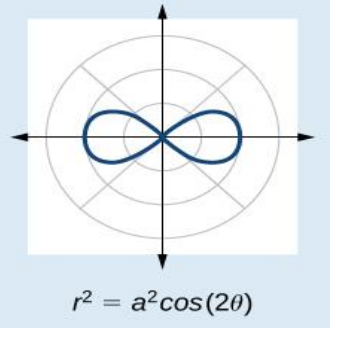
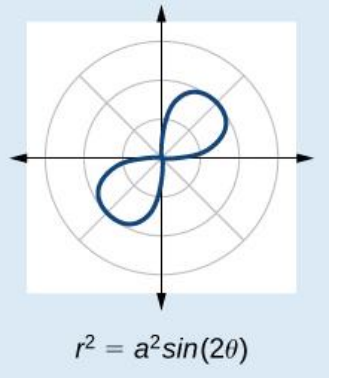
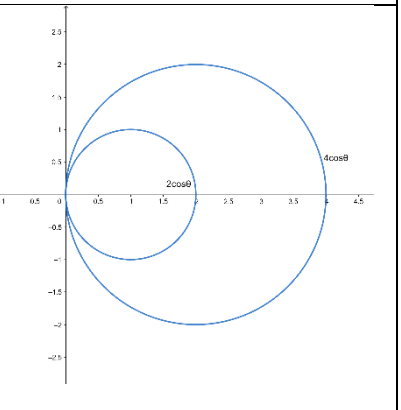
8	Cardioid $r = a(1-\sin\theta)$ $0 \le \theta \le \pi$ $0 \le r \le 2a$	
9	Bernoulli's lemniscate $r^2 = a^2 \cos 2\theta$	
10	Bernoulli's lemniscate $r^2 = a^2 \sin 2\theta$	
11	Region between $r = 2\cos\theta$ and $r = 4\cos\theta$	

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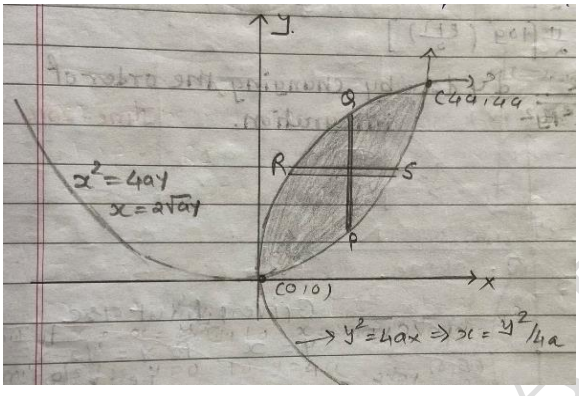
Cartesian Coordinates:

Example 1: Find the area between the parabolas $y^2 = 4ax$ & $x^2 = 4ay$.

Solution:



→ The points of intersection of $x^2 = 4ay$ and $y^2 = 4ax$ are obtained as

$(x^2)^2 = (4ay)^2 \Longrightarrow x^4 = 16a^2y^2$

$\Rightarrow x^4 = 16a^2(4ax)$

$\Rightarrow x^4 - 64a^3x = 0$

$\Rightarrow x(x^3 - 64a^3) = 0$

$\Rightarrow x = 0, x = 4a \Rightarrow y = 0, 4a$

The points of intersection are $(0,0)$ and $(4a, 4a)$.

→ Limits of $x : x = \dfrac{y^2}{4a}$ to $x = 2\sqrt{ay}$

→ Limits of $y : y = 0$ to $y = 4a$

Hence, the given integral after change of order is

$$\text{Area} = \int_0^{4a} \int_{\frac{y^2}{4a}}^{2\sqrt{ay}} dx\, dy$$

$$= \int_0^{4a} \left[\int_{\frac{y^2}{4a}}^{2\sqrt{ay}} dx \right] dy$$

$$= \int_0^{4a} [x]_{\frac{y^2}{4a}}^{2\sqrt{ay}} dy$$

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$$= \int_0^{4a} \left[2\sqrt{ay} - \frac{y^2}{4a} \right] dy$$

$$= \left[2\sqrt{a} \frac{y^{\frac{3}{2}}}{\frac{3}{2}} - \frac{1}{4a} \frac{y^3}{3} \right]_0^{4a}$$

$$= \left[\frac{4}{3}\sqrt{a}(4a)^{\frac{3}{2}} - \frac{1}{12a}(4a)^3 \right] - [0]$$

$$= \frac{32}{3}a^2 - \frac{16}{3}a^2$$

$$= \frac{16}{3}a^2$$

H.W.

Example 2: Find the propeller shaped region enclosed by the curves $x - y^{\frac{1}{3}} = 0$ & $x - y^{\frac{1}{5}} = 0$

Ans: $\frac{1}{6}$

Polar Coordinates:

Example 1: Find the area of the region that lies inside the cardioid $r = 1 + \cos\theta$ and outside the circle $r = 1$.

Solution:

→ The points of intersection of the cardioid $r = 1 + \cos\theta$ and the circle $r = 1$ is

$1 = 1 + \cos\theta$

$\Rightarrow \cos\theta = 0$

$\Rightarrow \theta = \pm\frac{\pi}{2}$

→ The region is symmetric about the initial line $\theta = 0$.

Total area = 2 (area above the initial line)

→ Limits of $r : r = 1$ to $r = 1 + \cos\theta$

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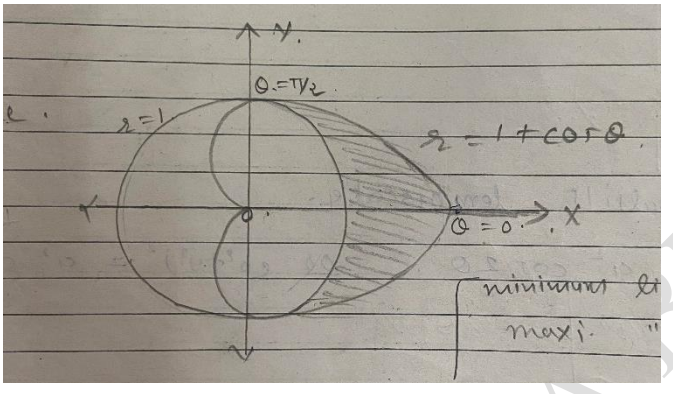
$\rightarrow$ Limits of $\theta : \theta = 0$ to $\theta = \dfrac{\pi}{2}$



Hence, required area

$$= 2\int_0^{\frac{\pi}{2}} \int_1^{1+\cos\theta} r\,dr\,d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \left[\int_1^{1+\cos\theta} r\,dr \right] d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \left[\int_1^{1+\cos\theta} \frac{r^2}{2}\,dr \right] d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left[(1+\cos\theta)^2 - 1 \right] d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left[1 + 2\cos\theta + \cos^2\theta - 1 \right] d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \cos\theta\,d\theta + \int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta$$

$$= 2\left[\sin\theta\right]_0^{\frac{\pi}{2}} + \frac{1}{2}\cdot\frac{\pi}{2}$$

$$= 2\left[\sin\frac{\pi}{2} - \sin 0 \right] + \frac{\pi}{4}$$

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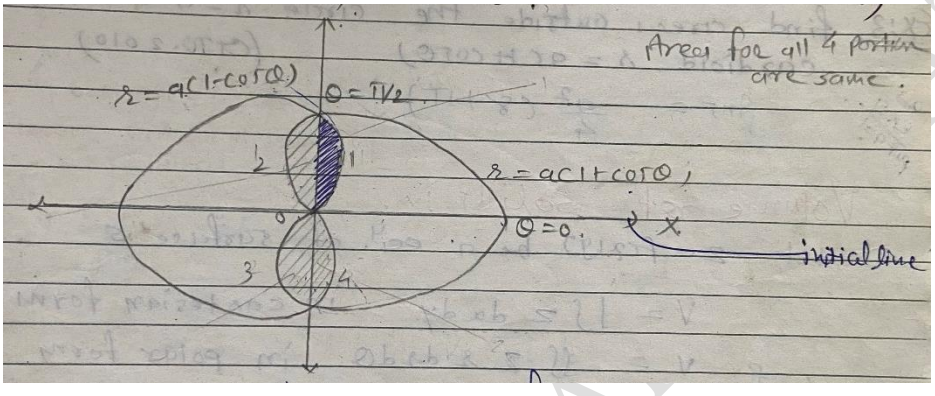
$$= 2(1) + \frac{\pi}{4}$$

$$= \frac{8+\pi}{4}$$

Example 2: Find the area common to the cardioids $r = a(1-\cos\theta)$ and $r = a(1+\cos\theta)$.

Solution:



→ The points of intersection of the cardioid $r = a(1-\cos\theta)$ and the circle $r = a(1+\cos\theta)$ is

$$a(1-\cos\theta) = a(1+\cos\theta)$$

$$\Rightarrow 1-\cos\theta = 1+\cos\theta$$

$$\Rightarrow 2\cos\theta = 0$$

$$\Rightarrow \cos\theta = 0$$

$$\Rightarrow \theta = \pm\frac{\pi}{2}$$

→ The region is symmetric in all the quadrants.
Total area = 4 (area above the initial line)

→ Limits of $r : r = 0$ to $r = a(1-\cos\theta)$

→ Limits of $\theta : \theta = 0$ to $\theta = \frac{\pi}{2}$

Hence, required area

$$= 4\int_0^{\frac{\pi}{2}} \int_0^{a(1-\cos\theta)} r\,dr\,d\theta$$

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$$= 4\int_0^{\frac{\pi}{2}} \left[\int_0^{a(1-\cos\theta)} r\,dr \right] d\theta$$

$$= 4\int_0^{\frac{\pi}{2}} \left[\frac{r^2}{2} \right]_0^{a(1-\cos\theta)} d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \left[\left(a(1-\cos\theta)\right)^2 - 0 \right] d\theta$$

$$= 2a^2 \int_0^{\frac{\pi}{2}} (1-\cos\theta)^2\, d\theta$$

$$= 2a^2 \int_0^{\frac{\pi}{2}} \left(1 - 2\cos\theta + \cos^2\theta\right) d\theta$$

$$= 2a^2 \left(\int_0^{\frac{\pi}{2}} d\theta - 2\int_0^{\frac{\pi}{2}} \cos\theta d\theta + \int_0^{\frac{\pi}{2}} \cos^2\theta d\theta \right)$$

$$= 2a^2 \left([\theta]_0^{\frac{\pi}{2}} - 2[\sin\theta]_0^{\frac{\pi}{2}} + \frac{1}{2}\frac{\pi}{2} \right)$$

$$= 2a^2 \left(\left[\frac{\pi}{2} - 0\right] - 2\left[\sin\frac{\pi}{2} - \sin 0\right] + \frac{1}{2}\frac{\pi}{2} \right)$$

$$= 2a^2 \left(\frac{\pi}{2} - 2[1-0] + \frac{1}{2}\frac{\pi}{2} \right)$$

$$= 2a^2 \left(\frac{3}{2}\frac{\pi}{2} - 2 \right)$$

$$= 2a^2 \left(\frac{3\pi - 8}{4} \right)$$

$$= \frac{(3\pi - 8)a^2}{2}$$

Example 3: Find the area lying inside the circle $r = a\sin\theta$ and outside the cardioid $r = a(1-\cos\theta)$.

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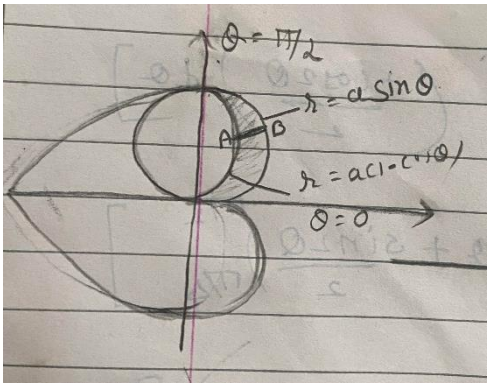
Solution:



→ The points of intersection of the circle $r = a\sin\theta$ and the cardioid $r = a(1-\cos\theta)$ is

$$a\sin\theta = a(1-\cos\theta)$$

$$\Rightarrow \sin\theta = 1-\cos\theta$$

$$\Rightarrow 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2\sin^2\frac{\theta}{2}$$

$$\Rightarrow \sin\frac{\theta}{2} = 0, \tan\frac{\theta}{2} = 1$$

$$\Rightarrow \frac{\theta}{2} = 0, \frac{\theta}{2} = \frac{\pi}{4}$$

$$\Rightarrow \theta = 0, \theta = \frac{\pi}{2}$$

→ Limits of $r : r = a(1-\cos\theta)$ to $r = a\sin\theta$

→ Limits of $\theta : \theta = 0$ to $\theta = \frac{\pi}{2}$

Hence, required area

$$= \int_0^{\frac{\pi}{2}} \int_{a(1-\cos\theta)}^{a\sin\theta} r\,dr\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left[\int_{a(1-\cos\theta)}^{a\sin\theta} r\,dr \right] d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left[\frac{r^2}{2} \right]_{a(1-\cos\theta)}^{a\sin\theta} d\theta$$

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$$=\frac{1}{2}\int_0^{\frac{\pi}{2}}\left[(a\sin\theta)^2-\left(a(1-\cos\theta)\right)^2\right]d\theta$$

$$=\frac{a^2}{2}\int_0^{\frac{\pi}{2}}\left[\sin^2\theta-\left(1-2\cos\theta+\cos^2\theta\right)\right]d\theta$$

$$=\frac{a^2}{2}\int_0^{\frac{\pi}{2}}\left[\sin^2\theta-1+2\cos\theta-\cos^2\theta\right]d\theta$$

$$=\frac{a^2}{2}\left(\int_0^{\frac{\pi}{2}}\sin^2\theta d\theta-\int_0^{\frac{\pi}{2}}d\theta+2\int_0^{\frac{\pi}{2}}\cos\theta d\theta-\int_0^{\frac{\pi}{2}}\cos^2\theta d\theta\right)$$

$$=\frac{a^2}{2}\left(\frac{1}{2}\frac{\pi}{2}-[\theta]_0^{\frac{\pi}{2}}+2[\sin\theta]_0^{\frac{\pi}{2}}-\frac{1}{2}\frac{\pi}{2}\right)$$

$$=\frac{a^2}{2}\left(-\left[\frac{\pi}{2}-0\right]+2\left[\sin\frac{\pi}{2}-\sin 0\right]\right)$$

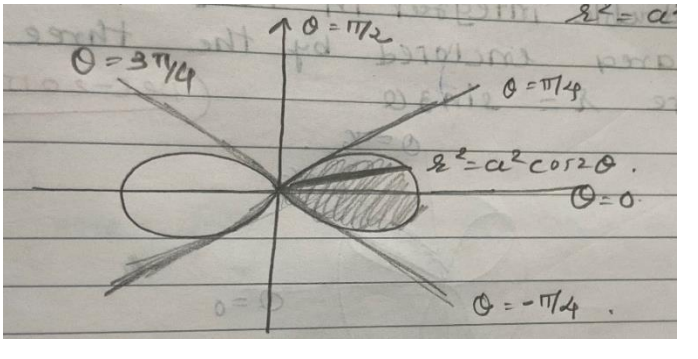
$$=\frac{a^2}{2}\left(-\frac{\pi}{2}+2[1-0]\right)$$

$$=\frac{a^2}{2}\left(2-\frac{\pi}{2}\right)$$

$$=a^2\left(1-\frac{\pi}{4}\right)$$

Example 4: Find the total area enclosed by lemniscate $r^2=a^2\cos 2\theta$.

Solution:



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→ The region is symmetric in all the quadrants.
Total area = 4 (area above the initial line)

→ Limits of $r : r = 0$ to $r = a\sqrt{\cos 2\theta}$

→ Limits of $\theta : \theta = 0$ to $\theta = \frac{\pi}{4}$

Hence, required area

$$= 4\int_0^{\frac{\pi}{4}} \int_0^{a\sqrt{\cos 2\theta}} r\,dr\,d\theta$$

$$= 4\int_0^{\frac{\pi}{4}} \left[\int_0^{a\sqrt{\cos 2\theta}} r\,dr \right] d\theta$$

$$= 4\int_0^{\frac{\pi}{4}} \left[\frac{r^2}{2} \right]_0^{a\sqrt{\cos 2\theta}} d\theta$$

$$= 2\int_0^{\frac{\pi}{4}} \left[\left(a\sqrt{\cos 2\theta} \right)^2 - 0 \right] d\theta$$

$$= 2a^2 \int_0^{\frac{\pi}{4}} \cos 2\theta\, d\theta$$

$$= 2a^2 \left[\frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{4}}$$

$$= a^2 \left[\sin\frac{\pi}{2} - \sin 0 \right]$$

$$= a^2[1 - 0]$$

$$= a^2$$

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Example 5: Evaluate $\iint_R r^3 \sin 2\theta dr d\theta$ over the area bounded in the first quadrant between circle $r = 2$ and $r = 4$.
Ans: 60

Example 6: Find the area common to the both circles $r = \cos\theta$ and $r = \sin\theta$.
Ans: $\frac{\pi}{8} - \frac{1}{4}$

Example 7: Find the area between to the both circles $r = 2\cos\theta$ and $r = 4\cos\theta$.
Ans: 3π

Example 8: Evaluate $\iint_R \frac{r}{\sqrt{r^2 + a^2}} dr d\theta$ over one loop of the lemniscate $r^2 = a^2 \cos 2\theta$.

Ans: $\frac{a}{2}(4 - \pi)$ `

Reduction Formula:

(1) $\int_0^{\frac{\pi}{2}} sin^m x . cos^n x \, dx = \left[\frac{\{(m-1)(m-3)(m-5).....2 \; or \; 1\}\{(n-1)(n-3)(n-5).....2 \; or \; 1\}}{(m+n)(m+n-2)(m+n-4).....2 \; or 1}\right] . k$

$; k = \frac{\pi}{2} \; if \; m = n = even$
$= 1 \; if \; otherwise$

(2) $\int_0^{\frac{\pi}{2}} sin^n x \, dx$ *or* $\int_0^{\frac{\pi}{2}} cos^n x \, dx = \frac{n-1}{n} . \frac{n-3}{n-2} . \frac{n-5}{n-4} \dots\dots\dots . \frac{1}{2} . \frac{\pi}{2} \; ; \; n = even$

$= \frac{n-1}{n} . \frac{n-3}{n-2} . \frac{n-5}{n-4} \dots\dots\dots . \frac{2}{3} . 1 \; ; n = odd$

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